



A new yield surface model considering non-uniform strain rate hardening behavior

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ABSTRACT

Uniaxial tensile tests using specimens of different geometries and stress states were performed on H340 steel sheets with various specimen orientations to rolling direction (RD) under strain rates ranging from 10^{-4} to 10^3 s^{-1} . Such uniaxial tensile tests, especially simple shear tests, show clear differences in work hardening that are significantly dependent on strain rate. This phenomenon, referred to as non-uniform strain rate hardening, has not been considered in existing yield surface models. Thus, a new yield surface model, termed the YldSRH model, based on Tsai-Wu strength criterion was proposed in this study to describe the non-uniform strain rate-dependent expansion of the yield surface. All parameters in the YldSRH model are independent of each other and can be directly determined using experimental results from tensile tests conducted in different directions and stress states at various strain rates. Moreover, the YldSRH model was coupled with a rate- and temperature-dependent damage mechanics (e^2 MBW) model for a combined YldSRH+ e^2 MBW model. The predictive capability of the YldSRH+ e^2 MBW model was validated by comparing the experimental and simulated force-displacement responses of different fracture specimens (SH, CH-R3, NDB-R8 and PS-R4) corresponding to different stress states at the three material orientations and at loading speeds of up to 10000 mm/s. In this study, the new model exhibited improved predictability for the dynamic anisotropic plasticity and fracture behavior of H340 compared with the existing model.

1. Introduction

Constitutive models are essential for precisely characterizing the plastic flow behaviors of sheet metals in the modern automotive industry. Although the conventional von Mises yield criterion coupled with isotropic hardening model is commonly employed, numerous advanced constitutive models have been developed over the past few decades to enhance plasticity prediction accuracy based on classic yield surface models such as the Hill [1–4], the Barlat [5–10], the Banabic [11–13] and other families [14–18].

For anisotropic materials, the yielding properties are direction-dependent, so the expression of the yield criterion may vary according to the reference frame. Hill [1] proposed the first quadratic anisotropic plasticity model by extending the isotropic von Mises criterion to orthotropic anisotropic materials with six anisotropy coefficients. The simplicity of the mathematical model and the low cost of the experiment have led to the widespread application and further development of the model [19,20]. In addition, Hill has proposed several non-quadratic yield functions [2–4] to address certain limitations such as considering

shear stress, capturing both first and second anomalous behaviors, and being user-friendly. Independent of Hill, Hosford [21] also proposed a non-quadratic yield criterion for materials that exhibit orthotropic symmetry. An important drawback of the Hill79 and Hosford79 yield criteria is that they do not involve shear stresses. Hence, in the Yld89 yield criterion, Barlat and Lian [5] extended Hosford79 criterion to account for the influence of shear stress. Subsequently, Barlat and co-workers proposed a series of anisotropic yield criteria to describe the plastic deformation behavior of sheet metals, including Yld91 [6], Yld2000-2d [8], Yld2004-18p [9], and Yld2011-27p [10]. Although introducing more anisotropic coefficients (e.g., the linear transformation tensor of the Cauchy stress) improves the effectiveness and accuracy of the anisotropic yield criteria, the homogeneous expansion of the anisotropic yield surface with increasing strain (which only considers the initial yield stress and r-value) could not predict the anisotropic evolution caused by plastic deformation. Consequently, significant efforts have been made to enhance the yield criteria by introducing yield surface anisotropic hardening [16,22–28] and the non-associated flow rule [18,19,22,29]. Moreover, to better capture the complex mechanical response of sheet metals under various loading

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Nomenclature		
C	Elastic stiffness matrix	
$C_{i=1-3}^T$	Temperature effects $g(T)$ parameters	
$\bar{C}_{i=1-4}^{\bar{\epsilon}_0}$	Strain rate sensitivity $f(\bar{\epsilon}_p)$ parameters	
D	Damage variable	
$D_{i=1-4}$	Damage initiation strain $\bar{\epsilon}_{ddi}^p$ parameters	
$F_{i=1-5}^{df}, T_m, T_0, \chi$	Ductile fracture strain $\bar{\epsilon}_{df}^p$ parameters	
$F_{i=1-6}, F_{ij=1-6}$	Anisotropic parameters of the YldSRH model	
$F_\sigma, G_\sigma, H_\sigma, L_\sigma, M_\sigma$ and N_σ	Six anisotropic parameters of the stress-based Hill48 model	
f	Yield criterion	
$f(\bar{\epsilon}_p)$	Strain rate sensitivity	
$g(T)$	Temperature effects	
I_{dd}	Ductile damage indicator	
m, D_{cr}, G_f	Parameters of damage accumulation rate, critical damage value, and fracture energy	
T	Temperature	
		$U_{t,0^\circ}, U_{t,45^\circ}, U_{t,90^\circ}, S_{0^\circ}, S_{45^\circ}, B_{t,xy}$ Yield strength in tension, shear and equi-biaxial tension
		β Taylor-Quinney coefficient
		$\beta_i, S_c, \bar{\epsilon}_c$ Taylor-Quinney coefficient β parameters
		ϵ^e Elastic strain tensor
		$\bar{\epsilon}^p$ Equivalent plastic strain
		$\dot{\bar{\epsilon}}_p$ Plastic strain rate
		$\bar{\epsilon}_{ddi}^p$ Ductile damage initiation strain
		$\bar{\epsilon}_{df}^p$ Ductile fracture strain
		η, η_{avg} Stress triaxiality and average stress triaxiality
		$\bar{\theta}, \bar{\theta}_{avg}$ Lode angle parameter and average Lode angle parameter
		σ Cauchy stress tensor
		$\bar{\sigma}$ Equivalent stress
		σ_{ref} Reference flow stress
		$\sigma_y(\bar{\epsilon}^p), \alpha, A, \bar{\epsilon}_0, n, k_0, Q, \beta$ Flow curve at the QS, RT and associated parameters
		$\chi_{i=1-4}$ χ parameters

conditions, more constitutive models have been developed to address different aspects of plasticity behavior, such as stress state [15,27,30,31], temperature [20,32–34], and microstructural evolution [35–39].

Most of the aforementioned models have been developed on the basis of traditional metal forming processes, where low or intermediate strain rates are typical. However, in the event of a vehicle crash, significant multiaxial deformation and failure occur at strain rates ranging from 10^{-4} to 10^3 s^{-1} [40–43]. Therefore, it is essential to develop a strain rate dependent phenomenological constitutive model that can accurately predict plasticity behavior under high strain rate loading conditions. Most dynamic phenomenological constitutive models proposed in the literature include yield surface models that consider only isotropic strain rate hardening [44–50]; that is, the increase in flow stress with increasing strain rate is the same in all stress states and material directions. These dynamic constitutive models include the multiplicative effects of strain rate and adiabatic temperature, typically focusing only on the RD and assuming that the strain rate sensitivity is isotropic.

However, evidence from dynamic tests [51–53] indirectly suggests the possibility of a non-uniform expansion of the yield surface with plastic strain and strain rate. These studies involved quasi-static (QS) and dynamic tests conducted at speeds of up to 2.5 m/s on samples with different geometries representing different stress states. Owing to the system ringing effect of the testing systems [40–42,53], the yield point of the materials could not be determined. However, the failure strain shows a strain rate dependence, which is strongly dependent on stress triaxiality and varies from material to material. This suggests that the strain rate dependency of material flow stress may also be stress state dependent, significantly influencing damage evolution and subsequent fracture behavior.

Thus, the precise measurement of force signals under high-speed loading is essential to evaluate the existence of the abovementioned effect. In the past two decades, many efforts have been made to reduce oscillations in servo-hydraulic machines by either developing new load sensor systems or designing specimen geometries and placing strain gauges in the second shoulder area of the specimens [40–42,54–58]. A new specimen geometry, known as the Gen. III specimen, along with a corresponding high-speed testing approach, was developed by Fang and Grams [56] to mitigate the system ringing effect in a servo-hydraulic machine (Zwick HTM5020). Compared with standard tensile specimens, this specimen has an additional second shoulder and a minor plastic deformation area (MinDA). An oscillation-free force can be measured in an elastic measurement area (EMA) in the second shoulder

using strain gauges at a strain rate of 1000 s^{-1} . For such kind of specimens with complex geometries, oscillation-free force can be measured under dynamic loading of up to 20 m/s ($> 1000 \text{ s}^{-1}$) by slightly modifying the details of the geometries [57,59–61]. Both finite element (FE) simulations and real tests have demonstrated that by using the Gen. III specimen, oscillation-free force-time curves could be obtained during tensile tests at strain rates ranging from QS to 1000 s^{-1} . On the basis of the oscillation-free force-displacement curves obtained at both low and high strain rates, the strain rate sensitivity of materials can be determined by performing various mechanical tests associated with specific stress states, such as uniaxial tensile tests in different loading directions (longitudinal, transverse, and diagonal) relative to the RD and simple shear tests [62]. As the strength of metals generally increases with increasing strain rate, this behavior is referred to as strain rate hardening in this work. The results of our experimental studies showed indications that strain rate hardening may not be uniform across loading directions or stress states, which is referred to as non-uniform strain rate hardening. However, no systematic investigation of this phenomenon has been conducted.

In addition to material anisotropic yielding and hardening, damage and fracture have also been extensively investigated and considered in many phenomenological constitutive models, such as the Johnson-Cook [44], CrachFEM [63], Bai-Wierzbicki (B-W) [30], Hosford-Coulomb [64], and GISSMO [65]. Lian et al. [66] proposed a damage mechanics model, referred to as the MBW model, based on the B-W model [30], which accounts for damage initiation and evolution in modeling the plasticity and ductile damage behavior of dual-phase steel sheets under QS loading conditions. The MBW model has been further adopted and enhanced (e.g., the eMBW model) by many researchers [47,61,67–69]. Building on this, Zeng et al. [61] developed a rate- and temperature-dependent damage mechanics model (e²MBW) to account for the comprehensive influences of strain rate and temperature on the damage and fracture behavior of sheet metals under dynamic loading conditions. However, the strain-rate sensitivity was assumed to be isotropic in all the cases. In other words, strain rate hardening was assumed to be uniform.

Therefore, the aim of the present study was to investigate the evolutions of material yielding and hardening with plastic strain at various strain rates and loading directions. It was also intended to evaluate the existence of anisotropy in strain rate sensitivity or the existence of the non-uniform strain rate hardening, and to develop a new yield surface model that can describe the behavior of such non-uniform strain rate

Table 1

Chemical composition of H340 (mass %).

Material	C	Si	Mn	Al	V	Nb	Ti
H340	0.03	0.13	0.73	0.040	0.001	0.010	0.001

hardening. Moreover, the yield surface model should be implemented into a user-subroutine and coupled with the previously developed e^2 MBW damage mechanics model to model material plasticity and fracture behaviors under dynamic loading conditions. The proposed model should also be evaluated to determine whether it improves the prediction of plasticity and fracture behavior.

This paper is structured as follows: Section 2 presents tensile tests with different specimen geometries (stress states) in 0° , 45° , and 90° loading directions relative to the RD and strain rates ranging from 10^{-4} to 10^3 s^{-1} on a high-strength micro-alloyed steel (H340). It also presents an evaluation of non-uniform strain rate hardening. In Section 3, the non-uniform strain rate-dependent yield surface model YldSRH is introduced, analyzed, and coupled with the e^2 MBW framework. Section 4 details the calibration and validation of the proposed anisotropic plasticity and damage mechanics model, emphasizing its predictive capabilities and improvements compared with existing models. Finally, the conclusions are summarized in Section 5.

2. Experimental setup and results

To investigate the yield strength and work hardening of materials in different loading directions at strain rates ranging from 10^{-4} to 10^3 s^{-1} , extensive experimental studies were conducted. For this purpose, the previously observed phenomenon that strain rate hardening behavior changes in different loading directions was systematically evaluated. This section outlines the investigated material and specimen geometries in Section 2.1, followed by a description of the experimental setup in Section 2.2. Section 2.3 will present the experimental results related to yielding, hardening, and fracture.

2.1. Material and specimen geometries

In this work, the investigated material (a micro-alloyed H340 steel) was delivered in the form of 1.5-mm-thick sheets. The chemical compositions of H340 are shown in Table 1.

Uniaxial tensile tests using specimens of various geometries (see Fig. 1a–b), which were developed in previous studies [56,57], were performed on H340 steel sheets under strain rates ranging from 10^{-4} to 10^3 s^{-1} . The Gen. III smooth dog bone (SDB) specimen (Fig. 1a), which was developed by Fang and Grams [56] has a section length of 20 mm and can realize a nominal strain rate of 1000 s^{-1} with a loading speed of 20 m/s by using the Zwick HTM5020 servo-hydraulic tensile machine. The other specimen geometries, namely SH, CH-R3, NDB-R8 and PS-R4 (shown in Fig. 1b), were designed and optimized according to the same physical principle used for the Gen. III specimen [55,57]. These geometries include much shorter gauge sections than those in Gen. III SDB specimen and can achieve an average local strain rate of higher than 1000 s^{-1} at a loading speed of 10 m/s for different stress states. The red dots represent the location of the virtual extensometers used to measure the relative displacements and speed using the digital image correlation (DIC) technique. These types of Gen. III specimens feature an additional second shoulder and a MinDA, enabling oscillation-free force measurement in an EMA using strain gauges at a strain rate of up to 1000 s^{-1} .

Uniaxial tension (SDB) tests at 0° , 45° , and 90° relative to the RD of the steel sheet, and simple shear (SH) tests at 0° were performed to

determine the non-uniform strain rate-dependent yield criterion. Details of these tests are described in Section 3.1.2. Tensile tests using four fracture specimens (see Fig. 1b) at various material orientations and loading speeds were performed to calibrate and subsequently validate the model parameters. Details are discussed in Section 4.

2.2. Experimental setup

In this work, tensile tests at low strain rates ($0.001 - 0.1 \text{ s}^{-1}$) and low loading speeds (0.005 and 0.5 mm/s) were performed on a 100-kN universal tensile testing machine (Zwick Z100). The force was directly measured with the load cell. For the high-speed tensile tests, a servo-hydraulic tensile machine (Zwick HTM 5020) was employed, as shown in Fig. 1c. The maximum loading velocity of HTM 5020 was 20 m/s. For strain rates higher 100 s^{-1} or impact loading speeds higher than 2 m/s, the force was measured in the EMA region of the specimen shoulder using strain gauges instead of the load cell, as shown in Fig. 1d (compared with Fig. 1a).

The displacement and strain information of the specimens during tensile testing were obtained using DIC methods. Two GOM ARAMIS 5 M CCD cameras with a maximum frame rate of 29 Hz were used for the tensile tests at low strain rates. For the high-speed tensile tests, two PHOTRON SA5 high-speed cameras with a maximum frame rate of 1 MHz were used to capture the images during specimen deformation and fracture (see Fig. 1d). Table 2 summarizes the DIC parameters used for the displacement and strain calculations. A uniform reference length, l_0 , of approximately 0.24 mm was applied for the local strain evaluation across all specimens at various loading speeds.

2.3. Experimental results

2.3.1. Uniaxial tension and simple shear tests at different strain rates

The experimental true stress-true plastic strain curves of SDB specimens at room temperature (RT) and various strain rates ($0.001 - 1000 \text{ s}^{-1}$) are shown in Fig. 2a–c, corresponding to different loading directions (0° , 45° , and 90°). Three parallel tests were conducted for each loading condition. As the tensile tests for the investigated steel H340 had good repeatability, only one curve is shown in Fig. 2. H340 exhibited a very strong strain rate hardening behavior. Fig. 2d shows the true stress-true plastic strain curves of the 0° simple shear test specimen (SH) at nominal strain rates ranging from 0.0016 to 3200 s^{-1} , corresponding to loading speeds of 0.005 to 10000 mm/s.

For uniaxial tension at 0° , 45° , and 90° , the true stress-true plastic strain curves were derived from the engineering stress-strain curves up to the point of necking. For the test with the SH specimen, a work-conjugate equivalent strain, as proposed in the literature [70–72], was applied to evaluate the strain for the hardening curve. A detailed calculation process of the shear stress and equivalent strain can be found in Appendix A.

A comparison of the strain rate sensitivities of the 0° , 45° , and 90° tensile and 0° shear tests at a strain of 0.02 for H340 is shown in Fig. 2e.

The strain rate sensitivity $f(\bar{\epsilon}_p)$ was calculated by comparing the true stress at various strain rates with that at 0.001 s^{-1} , i.e., $f(\bar{\epsilon}_p) = \sigma_{\bar{\epsilon}_p} / \sigma_{0.001 \text{ s}^{-1}}$. It was observed that the strain rate sensitivities of H340 varied across the loading directions. The strain rate sensitivity for 0° shear was significantly higher than that for uniaxial tension. In addition, the strain rate sensitivity for the 45° tension was noticeably higher than those for the 0° and 90° uniaxial tensions. Thus, the existence of a significant non-uniform strain rate hardening in relation to RD

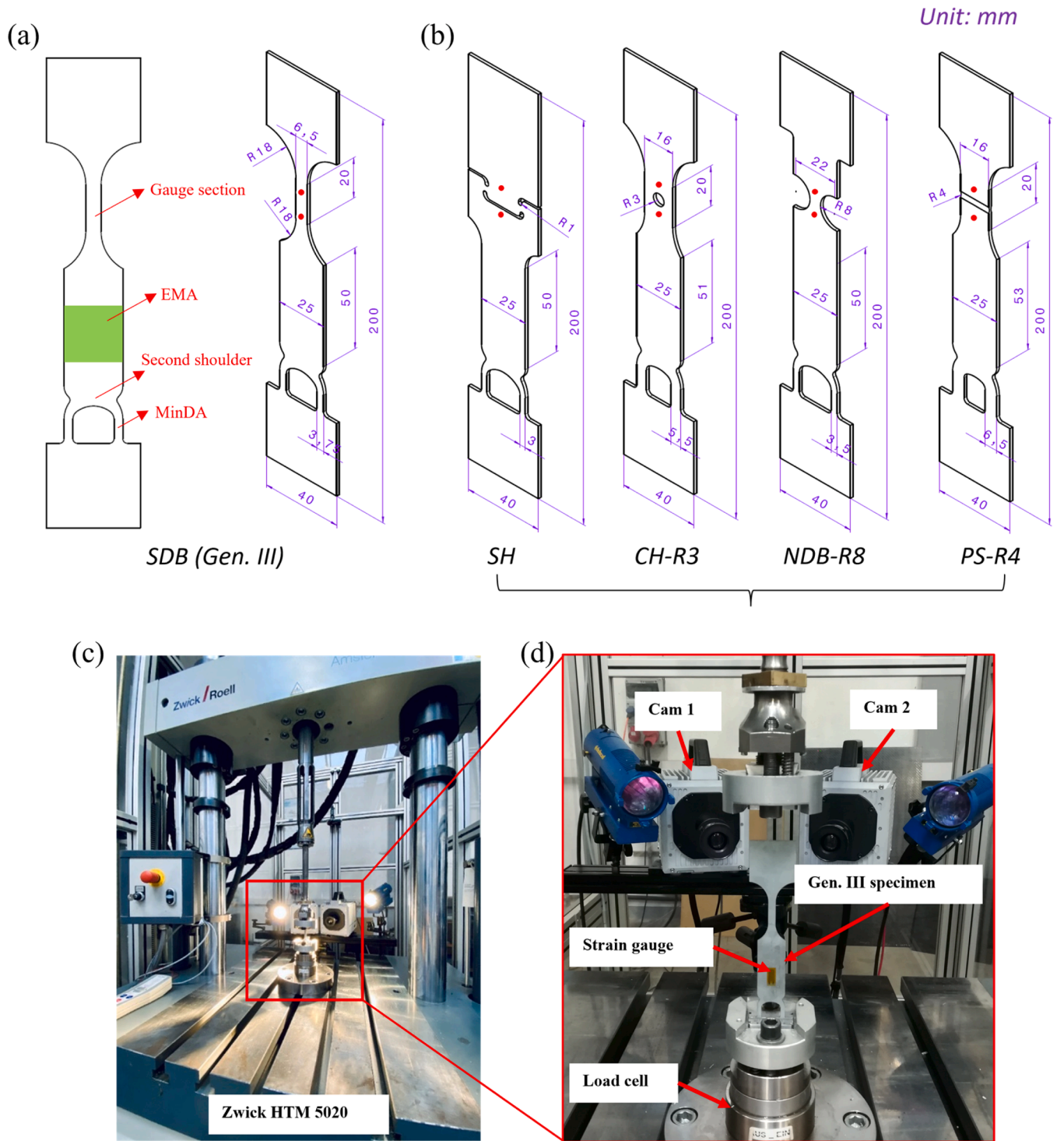


Fig. 1. Overview of the specimen geometries used for material characterization and the experimental setup for the high-speed tensile tests. (a-b) Gen. III specimen [56] and various geometries corresponding to different stress states ($\eta, \bar{\theta}$) [57]. The thickness of the sheet steel is 1.5 mm. (c) Zwick servo-hydraulic tensile machine with a maximum loading velocity of 20 m/s. (d) Details of the high-speed camera system and Gen. III specimen. The loading force was measured using strain gauge instead of load cell.

was confirmed.

In addition, strain rate sensitivity was found to be strain dependent. An example, based on the 45° tensile results, is presented in Fig. 2f, which shows the strain rate sensitivities at different strain rates and true

plastic strains. The experimental results indicate that as the true plastic strain increased from 0.02 to 0.12, the strain rate sensitivity had a trend of decreases. More regarding the strain rate sensitivity analysis and its fitting will be discussed in Section 4.1.3.

Table 2

Summary of the DIC (GOM Aramis) parameters under different loading speeds.

Loading speed, mm/s	Frame rate, Hz	Image scale, mm/pixel	Facet size, pixel	$D_{\text{facetpoints}}$ pixel	l_0 , mm
≤ 0.02	1	0.027	12	6	0.24
> 0.02 and ≤ 2	29	0.053	8	3	0.24
200	40000	0.054	8	3	0.24
2000	75000				
10000/20000	150000				

2.3.2. Fracture characterization at different loading speeds

Fig. 3 and Fig. 4 show the force-displacement curves of the different specimens (see Fig. 1b) in the three loading directions (0° , 45° , and 90°) under different loading speeds (0.005, 0.5, 200, 2000, and 10000 mm/s). A 10-mm gauge length was used to evaluate displacement in the DIC evaluation (see Fig. 1b). The force level of all four specimens in all loading directions exhibited a significant increasing trend as the loading speeds increased from 0.005 mm/s to 10000 mm/s.

Fig. 5 depicts the normalized forces of the different specimens at a given displacement for the three loading directions under low- (0.005 mm/s), intermediate- (200 mm/s) and high-speed (10000 mm/s) conditions. The given displacements of the CH-R3, NDB-R8, and PS-R4 specimens were approximately 1.1, 0.8, and 0.5 mm, respectively, corresponding to the displacements where maximum force occurred. For the SH specimen, a displacement of 0.2 mm, which is close to the initial yielding point, was chosen as an example. The effects of strain rate hardening are highly dependent on displacement (deformation) for shear loading conditions (see Fig. 3a-c). The results shown in Fig. 5 clearly demonstrate that the investigated H340 steel exhibited non-uniform strain rate hardening across all three loading speeds.

Fig. 5 also shows that at a low loading speed of 0.005 mm/s, the force (strength) increased as the loading orientation changed from 0° to 90° . However, at higher loading speeds of 200 and 10000 mm/s, the 45° loading direction exhibited mostly the highest force (strength), especially for the CH-R3, NDB-R8, and PS-R4 specimens. In addition, SH exhibited especially high strain rate dependency across all three directions. This indicates that the strain rate hardening of H340 steel is significantly influenced by the loading directions. Consequently, this resulted in a non-uniform evolution of the yield surface in the stress space at the different strain rates, which cannot be ignored, as has typically been the case. Thus, a new non-uniform strain rate-dependent yield surface model was proposed, as described in Section 3.1.2.

3. Constitutive model

In previous studies [44,46,47,61,73], strain rate sensitivity has been consistently determined from uniaxial tensile tests conducted in the material RD. To address the strain rate $\dot{\epsilon}_p$ and adiabatic temperature T effects, the yield function in the strain rate-dependent model (isotropic or anisotropic) is usually expressed as:

$$f = \bar{\sigma}(\alpha_{i-j}, \sigma) - \sigma_{ref}(\bar{\epsilon}^p) \cdot f(\bar{\epsilon}_p, T) \leq 0, \quad (1)$$

where $\bar{\sigma}$ is the equivalent stress and α_{i-j} are the anisotropy parameters [1,20,74] that correlate the response of the material under arbitrary loading conditions defined by the stress tensor σ with the reference stress state $\sigma_{ref}(\bar{\epsilon}^p)$. The reference flow stress σ_{ref} is often obtained using uniaxial tensile tests in the material RD (e.g., $\sigma_{ref} = \sigma_0$).

To account for the observed non-uniform strain rate hardening in the Section 2, this study proposed a new non-uniform strain rate-dependent yield surface model, as outlined in Section 3.1. This yield criterion was further coupled with the e^2 MBW damage mechanics model and implemented as a user-defined material model in LS-DYNA, which will be described in Section 3.2.

3.1. New non-uniform strain rate-dependent yield surface model

3.1.1. Strength criterion for anisotropic materials based on the Tsai-Wu criterion

A general theory of strength for brittle anisotropic materials was proposed by Tsai and Wu [75]. This theory was especially proposed for the failure prediction of anisotropic fiber reinforced materials that have different strengths under uniaxial tension, compression and pure shear loading conditions. The basic assumption of the strength criterion is that a failure surface exists in the stress space, which can be described with the following equation:

$$f(\sigma_k) = F_i \sigma_i + F_{ij} \sigma_i \sigma_j = 1, \quad (2)$$

where $i, j, k = 1, 2, \dots, 6$, and F_i and F_{ij} are strength tensors of the second and fourth ranks, respectively. This equation is automatically invariant, and each stress component shows an independent material property. Assuming symmetry in the stress tensor (i.e., $F_{ij} = F_{ji}$), the general form of the Tsai-Wu failure criterion is expressed as follows:

$$\left(\begin{aligned} &F_1 \sigma_1 + F_2 \sigma_2 + F_3 \sigma_3 + F_4 \sigma_4 + F_5 \sigma_5 + F_6 \sigma_6 \\ &+ F_{11} \sigma_1^2 + 2F_{12} \sigma_1 \sigma_2 + 2F_{13} \sigma_1 \sigma_3 + 2F_{14} \sigma_1 \sigma_4 + 2F_{15} \sigma_1 \sigma_5 + 2F_{16} \sigma_1 \sigma_6 \\ &+ F_{22} \sigma_2^2 + 2F_{23} \sigma_2 \sigma_3 + 2F_{24} \sigma_2 \sigma_4 + 2F_{25} \sigma_2 \sigma_5 + 2F_{26} \sigma_2 \sigma_6 \\ &+ F_{33} \sigma_3^2 + 2F_{34} \sigma_3 \sigma_4 + 2F_{35} \sigma_3 \sigma_5 + 2F_{36} \sigma_3 \sigma_6 \\ &+ F_{44} \sigma_4^2 + 2F_{45} \sigma_4 \sigma_5 + 2F_{46} \sigma_4 \sigma_6 \\ &+ F_{55} \sigma_5^2 + 2F_{56} \sigma_5 \sigma_6 \\ &+ F_{66} \sigma_6^2 \end{aligned} \right) = 1, \quad (3)$$

where the σ_1 , σ_2 , and σ_3 correspond to the normal stresses along the principal directions of the orthotropic reference axes (x, y, z), while the σ_4 , σ_5 , and σ_6 represent the shear stresses acting on the x-y, y-z, and x-z planes, respectively.

The Tsai-Wu failure criterion has 6 independent strength components for F_i and 21 for F_{ij} . Thus, F_i and F_{ij} can be written as follows:

$$F_i = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \\ F_6 \end{bmatrix}, \quad F_{ij} = \begin{bmatrix} F_{11} & F_{12} & F_{13} & F_{14} & F_{15} & F_{16} \\ F_{22} & F_{23} & F_{24} & F_{25} & F_{26} & \\ F_{33} & F_{34} & F_{35} & F_{36} & & \\ F_{44} & F_{45} & F_{46} & & & \\ F_{55} & F_{56} & & & & \\ F_{66} & & & & & \end{bmatrix}. \quad (4)$$

If a material exhibits some form of symmetry, certain interaction terms are expected to vanish. For orthotropic materials, the value of the terms F_4 , F_5 , and F_6 are expected to be zero [75]. In addition, the coupling between normal and shear strengths, such as $F_{14} - F_{16}$, $F_{24} - F_{26}$, $F_{34} - F_{36}$, will also vanish under the assumption that a change in the sign of shear stress does not affect the failure stress. Following similar reasoning, the shear strengths for an orthotropic material are assumed to be uncoupled, which means $F_{45} = F_{46} = F_{56} = 0$. However, the coupling between normal strengths is expected to remain, as proposed by Tsai and Wu [75]. With these symmetry assumptions, the numbers of independent strength components for F_i and F_{ij} are reduced to 3 and 9, respectively.

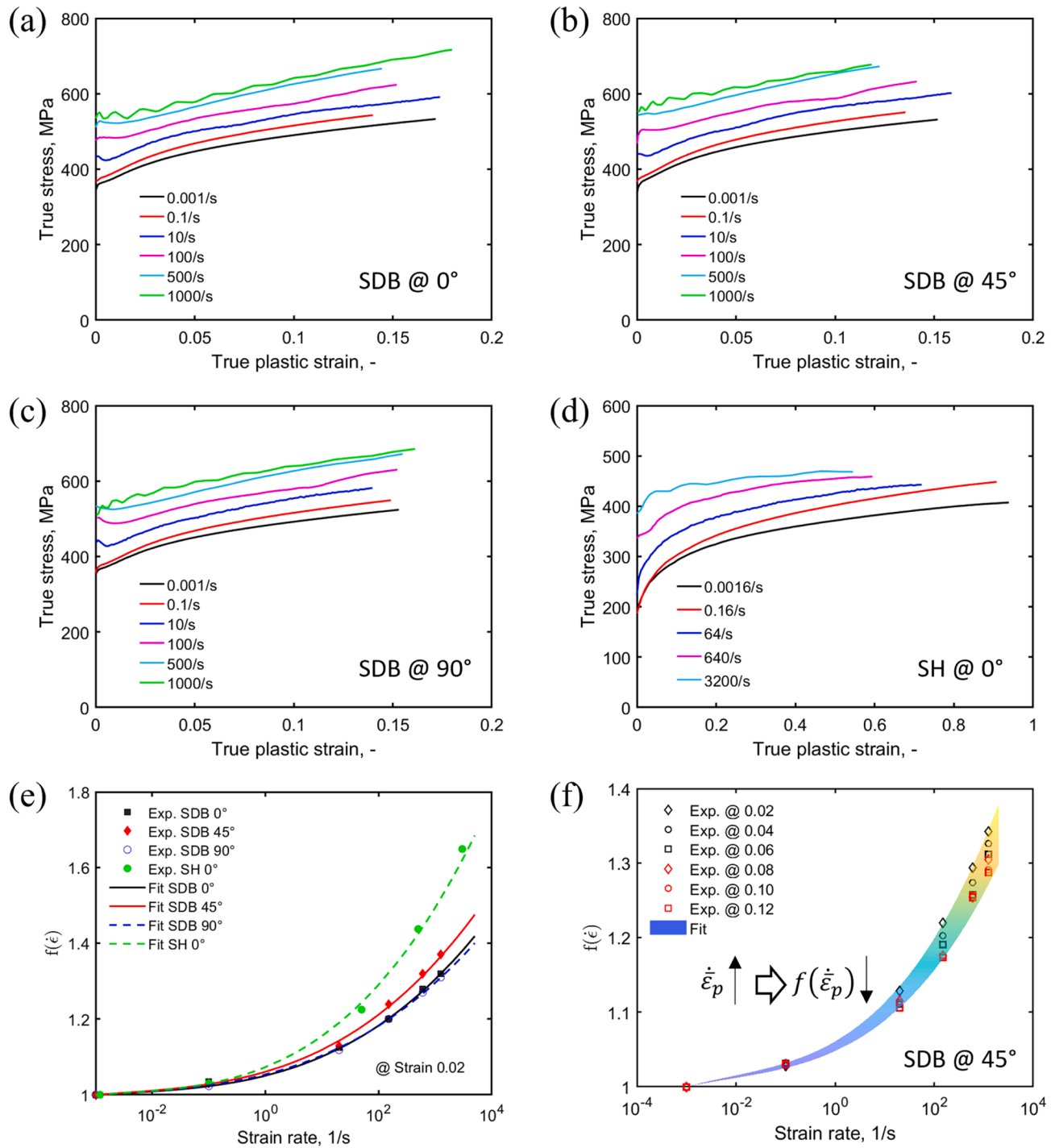


Fig. 2. True stress-true plastic strain curves and strain rate sensitivity analysis. (a-c) True stress-true plastic strain curves of SDB specimen for various strain rates ($0.001 - 1000 \text{ s}^{-1}$) in the 0° , 45° , and 90° directions. (d) True stress-true plastic strain curves of SH specimen for various strain rates ($0.0016 - 3200 \text{ s}^{-1}$). (e) Comparison of the strain rate sensitivities of the 0° , 45° , and 90° tensile and 0° shear tests at a strain of 0.02 for H340. (f) Strain rate sensitivities of the 45° tensile at different strain rates and true plastic strains.

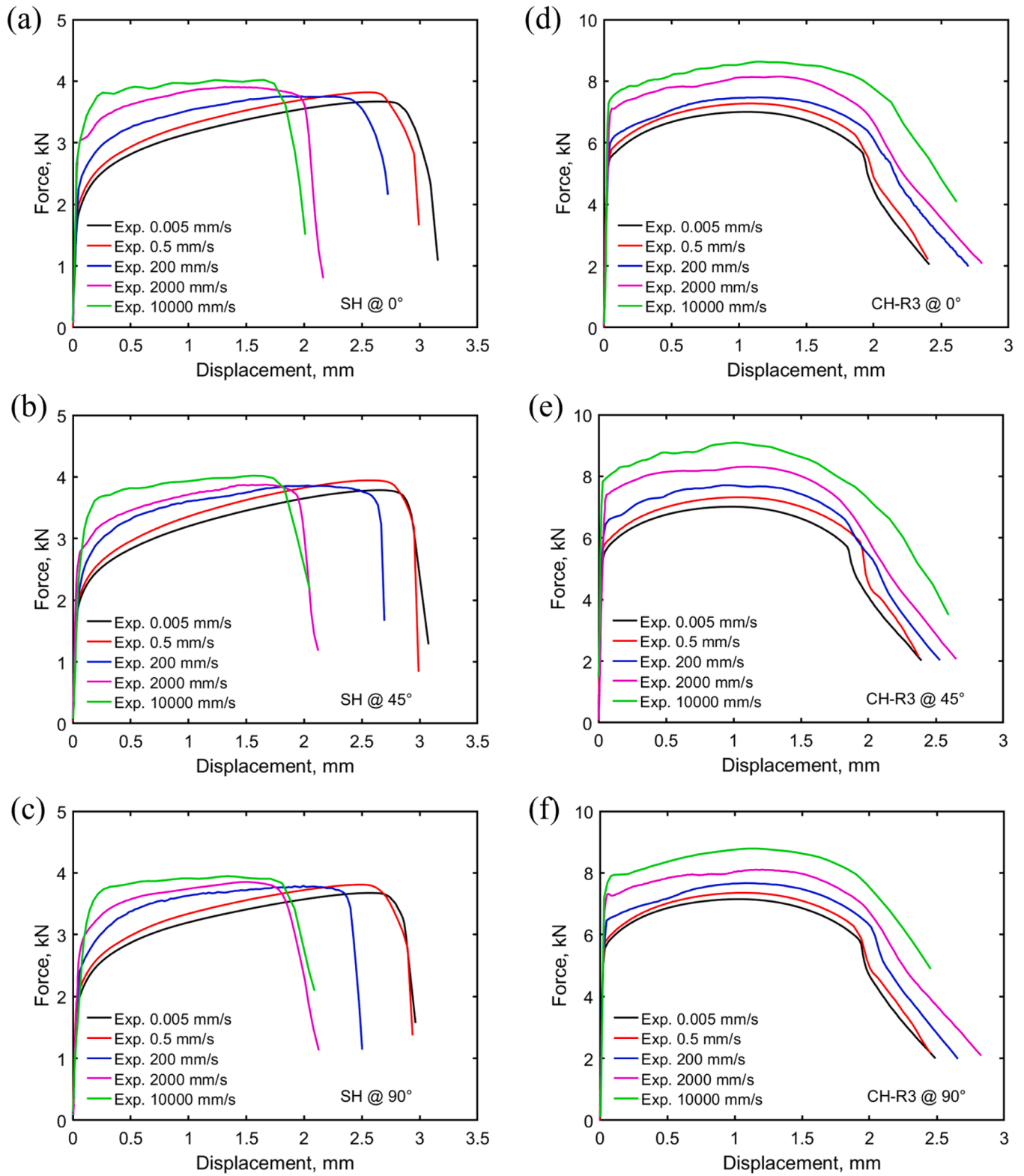


Fig. 3. Experimental force-displacement curves of H340 at different loading speeds (0.005 mm/s – 10000 mm/s) and material orientations (0°, 45°, and 90°). (a-c) SH specimen and (d-f) CH-R3 specimen.

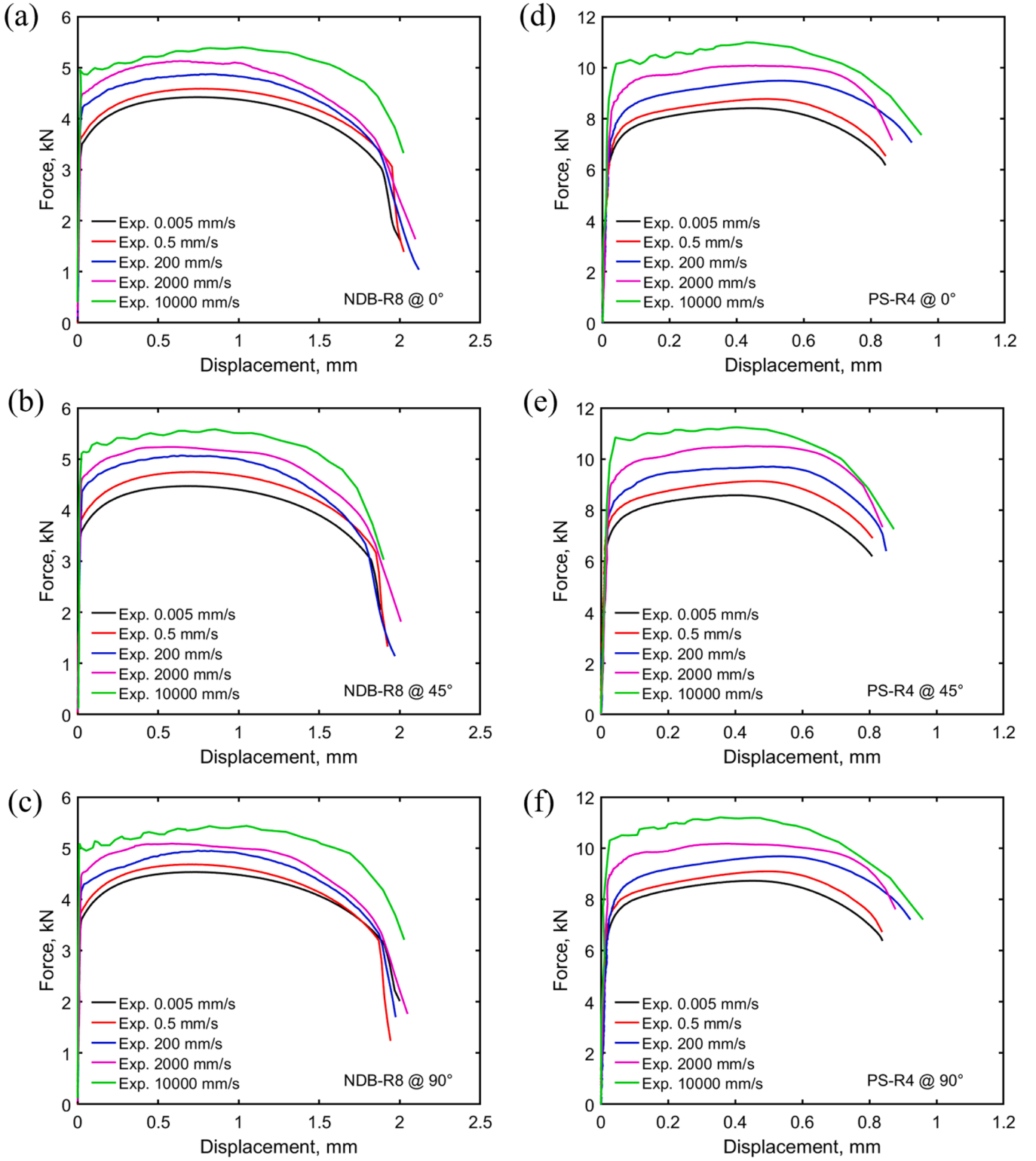


Fig. 4. Experimental force-displacement curves of H340 at different loading speeds (0.005 mm/s – 10000 mm/s) and material orientations (0°, 45°, and 90°). (a-c) NDB-R8 specimen and (d-f) PS-R4 specimen.

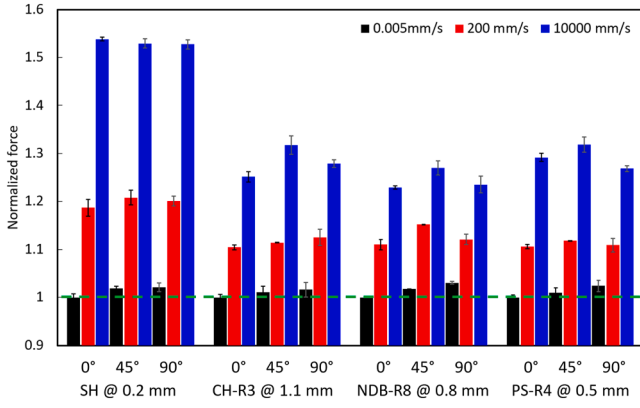


Fig. 5. Normalized force at different displacements for various specimens in the 0°, 45°, and 90° directions at different loading speeds.

$$F_i = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}, F_{ij} = \begin{bmatrix} F_{11} & F_{12} & F_{13} \\ F_{22} & F_{23} \\ F_{33} \\ F_{44} \\ F_{55} \\ F_{66} \end{bmatrix}. \quad (5)$$

The general form of the Tsai-Wu failure criterion is simplified as follows:

$$f(\sigma) = \left(\begin{array}{c} F_1\sigma_{11} + F_2\sigma_{22} + F_3\sigma_{33} \\ +F_{11}\sigma_{11}^2 + F_{22}\sigma_{22}^2 + F_{33}\sigma_{33}^2 + 2F_{12}\sigma_{11}\sigma_{22} + 2F_{13}\sigma_{11}\sigma_{33} + 2F_{23}\sigma_{22}\sigma_{33} \\ +F_{44}\sigma_{12}^2 + F_{55}\sigma_{23}^2 + F_{66}\sigma_{13}^2 \end{array} \right) = 1, \quad (6)$$

where the stress tensor components $\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5$, and σ_6 are replaced with $\sigma_{11}, \sigma_{22}, \sigma_{33}, \sigma_{12}, \sigma_{23}$, and σ_{13} for better understanding. The detailed determination process for parameters F_i and F_{ij} in Eq. (6) can be found in Appendix B.

3.1.2. New yield surface model

On the basis of the Tsai-Wu strength criterion for the failure of brittle anisotropic materials, a new yield function (YldSRH model) that adopts the failure surface $f(\sigma)$ according to Eq. (6) is proposed for describing the yield surface of ductile materials as follows:

$$f = (F_1\sigma_{11} + F_2\sigma_{22} + F_3\sigma_{33} + F_{11}\sigma_{11}^2 + F_{22}\sigma_{22}^2 + F_{33}\sigma_{33}^2 + 2F_{12}\sigma_{11}\sigma_{22} + 2F_{23}\sigma_{22}\sigma_{33} + 2F_{13}\sigma_{11}\sigma_{33} + F_{44}\sigma_{12}^2 + F_{55}\sigma_{23}^2 + F_{66}\sigma_{13}^2)^{\frac{1}{2}} \leq 1, \quad (7)$$

where F_i and F_{ij} are anisotropic parameters that evolve with the plastic strain $\bar{\epsilon}^p$ at different strain rates and temperatures. If internal stresses or Bauschinger's effects (kinematic hardening) are not considered, the F_i term can be omitted. In this case, similar to the Hill48 model [1], the yield function shown in Eq. (7) will be simplified to contain only quadratic terms. From the formulations for F_i and F_{ij} in Appendix B, it can be seen that all terms in F_i and F_{ij} can be directly determined from the experiment and are fully independent of each other (see Eqs. B3-B5, B8, and B9). By contrast, for the Hill48 model, the parameters F, G, H , and N interact with each other (see Eq. C2 in Appendix C).

It is worth mentioning that for metals subjected to complex loading conditions, such as tension-torsion, the interaction between normal and

shear stresses ($F_{14} - F_{16}, F_{24} - F_{26}$ and $F_{34} - F_{36}$) in Eq. (4) may no longer be ignored. However, since this work considers only uniaxial, shear, or biaxial loading, these interactions are negligible.

In this study, to consider material anisotropic hardening, the instantaneous values of yield stress at different strains $\bar{\epsilon}^p$ and strain rates $\dot{\bar{\epsilon}}_p$ were updated and used to calculate the anisotropic parameters of F_{ij} in Eq. (7). From a practical point of view only the parameters F_{11}, F_{22}, F_{44} and F_{12} in Eq. (7) are relevant in sheet metal forming and crashing analyses. The parameters F_{11}, F_{22} and F_{44} are defined as follows:

$$F_{11} = \frac{1}{U_{t,0^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)^2}; F_{22} = \frac{1}{U_{t,90^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)^2}; F_{44} = \frac{1}{S_{0^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)^2}, \quad (8)$$

where $U_{t,0^\circ}$ and $U_{t,90^\circ}$ are the material tensile strength in the 0° and 90° directions, and S_{0° is the in-plane 0° SH strength, under the assumption that the tensile and compression strengths are the same (see Appendix B, Eqs. B3-B5). F_{12} can be determined using tensile strength at 45°, equibiaxial tensile strength, or 45° off-axis pure shear strength, as shown in Eq. (9) – Eq. (11) [75] (see also Appendix B, Eqs. B8-B9). The terms defined in Eq. (9) – Eq. (11) must satisfy the inequality to ensure a definition of a closed surface.

$$F_{12} = \frac{2}{U_{t,45^\circ}^2(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)} \left[1 - \frac{U_{t,45^\circ}^2(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)}{4} (F_{11} + F_{22} + F_{44}) \right] < \pm \sqrt{F_{11}F_{22}}. \quad (9)$$

$$F_{12} = \frac{1}{2B_{t,xy}^2(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)} \left[1 - B_{t,xy}^2(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T) (F_{11} + F_{22}) \right] < \pm \sqrt{F_{11}F_{22}}. \quad (10)$$

$$F_{12} = -\frac{1}{2S_{45^\circ}^2(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)} \left[1 - S_{45^\circ}^2(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T) (F_{11} + F_{22}) \right] < \pm \sqrt{F_{11}F_{22}}. \quad (11)$$

Table 3

Yield stresses at different stress states for determining the yield criterion.

Yield criterion/parameter Determination type	Yield stresses required			
YldSRH Type 1 (Eq. (8) + (9))	$U_{t,0^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$U_{t,90^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$U_{t,45^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$S_{0^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$
YldSRH Type 2 (Eq. (8) + (10))	$U_{t,0^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$U_{t,90^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$B_{t,xy}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$S_{0^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$
YldSRH Type 3 (Eq. (8) + (11))	$U_{t,0^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$U_{t,90^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$S_{45^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$S_{0^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$
Hill48 model	$U_{t,0^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$U_{t,45^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$U_{t,90^\circ}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$	$B_{t,xy}(\bar{\epsilon}^p, \dot{\bar{\epsilon}}_p, T)$

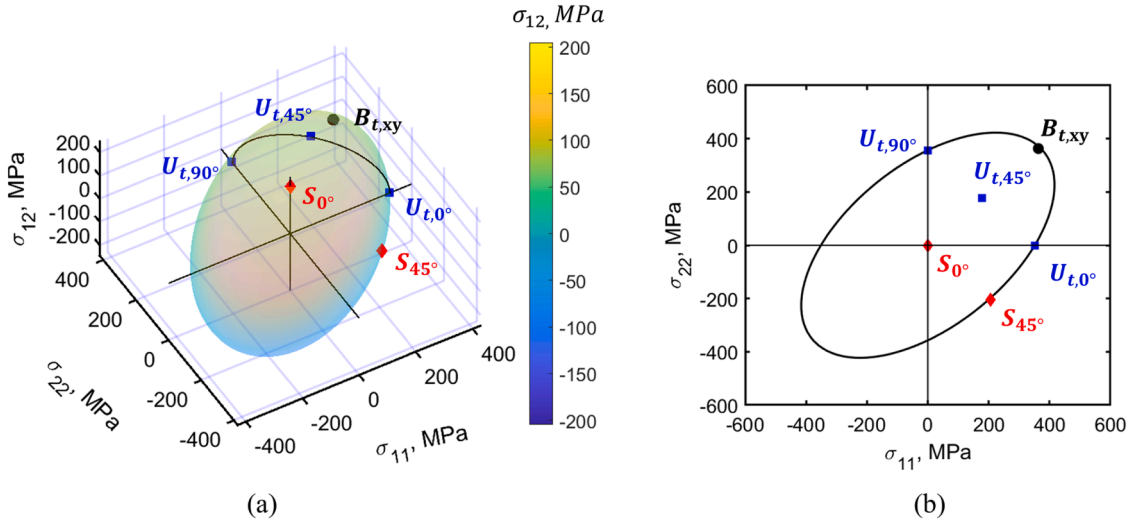


Fig. 6. Yield surface of H340 at a strain of 0.002 and strain rate of 0.001 s^{-1} . (a) Yield surface in the $\sigma_{11} - \sigma_{22} - \sigma_{12}$ stress space, (b) Yield locus at $\sigma_{12} = 0$.

Table 4

Yield stresses of H340 at a strain of 0.002 and strain rate of 0.001 s^{-1} for different stress states (MPa).

$U_{t,0^\circ}$	$U_{t,45^\circ}$	$U_{t,90^\circ}$	$B_{t,xy}$	S_{0°	S_{45°
352	358	369	363	207	207

The determination of the remaining parameters, namely F_{33} , F_{55} , F_{66} , F_{13} , and F_{23} , is quite challenging in mechanical tests because either tensile/compression tests in the normal direction or out-of-plane shear tests are difficult to realize in thin metal sheets. Alternative approaches, such as simulating out-of-plane tests using crystal plasticity models,

$$f = \left[F_{11}\sigma_{11}^2 + F_{22}\sigma_{22}^2 + \frac{4F_{11}F_{22}}{(\sqrt{F_{11}} + \sqrt{F_{22}})^2}\sigma_{33}^2 + 2F_{12}(\sigma_{11}\sigma_{22} + \sigma_{22}\sigma_{33} + \sigma_{11}\sigma_{33}) + F_{44}(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{13}^2) \right]^{\frac{1}{2}} \leq 1. \quad (12)$$

If $U_{t,0^\circ} = U_{t,45^\circ} = U_{t,90^\circ} = B_{t,xy} = \sqrt{3}S_{0^\circ} = \sqrt{3}S_{45^\circ}$, then the von Mises criterion is recovered (see Eq. (13)).

$$f = \frac{1}{U_{t,0^\circ}} (\sigma_{11}^2 + \sigma_{22}^2 + \sigma_{33}^2 + \sigma_{11}\sigma_{22} + \sigma_{11}\sigma_{33} + \sigma_{22}\sigma_{33} + 3\sigma_{12}^2 + 3\sigma_{23}^2 + 3\sigma_{13}^2)^{\frac{1}{2}} = \frac{1}{U_{t,0^\circ}} \left\{ \frac{1}{2} [(\sigma_{11} - \sigma_{22})^2 + (\sigma_{22} - \sigma_{33})^2 + (\sigma_{11} - \sigma_{33})^2] + 3(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{13}^2) \right\}^{\frac{1}{2}} = \frac{1}{U_{t,0^\circ}} \sigma_{\text{von Mises}} \leq 1. \quad (13)$$

could be explored to obtain flow curves under such loading conditions in future work [39,76–79]. In this work, F_{33} in Eq. (7) is defined under the assumption that the tensile strength in normal directions is the mean of that in longitudinal and transverse directions ($U_{t,normal} = (U_{t,0^\circ} + U_{t,90^\circ})/2$). The calculation of $U_{t,normal}$ was simplified due to the lack of accurate flow curves in the normal direction at different strain rates. The development of compression testing methods and equipment for accurately determining these flow curves at different strain rates, especially at high strain rate conditions, is essential and remains part of our ongoing investigation. F_{55} and F_{66} are often defined as isotropic parameters on the basis of von Mises' assumption (e.g., $S_{0^\circ} = \frac{U_{t,0^\circ}}{\sqrt{3}}$), as commonly used in Hill48 based models [19,80]. However, in this study, the values of F_{55} and F_{66} are considered equivalent to that of F_{44} . Analogously, F_{13} and F_{23} are simplified to be the same as F_{12} . These assumptions are reasonable because for thin sheet metals, the values of σ_{33} , σ_{13} , and σ_{23} are relatively small. Finally, for thin metal sheets without kinematic hardening effects ($F_{i=1\sim3}$), the yield function in Eq. (7) can be expressed as follows:

For plane stress conditions, Eq. (12) can be reduced to Eq. (14):

$$f = (F_{11}\sigma_{11}^2 + F_{22}\sigma_{22}^2 + 2F_{12}\sigma_{11}\sigma_{22} + F_{44}\sigma_{12}^2)^{\frac{1}{2}} \leq 1. \quad (14)$$

According to Eq. (14), at least four hardening curves for different stress states are required to determine the proposed evolving YldSRH yield criterion. There are three combinations based on different calculation methods for F_{12} (see Eq. (9) – Eq. (11)).

Table 3 shows the four yield stresses at different stress states that are required for determining the YldSRH yield criterion. It is obvious that for the proposed YldSRH yield criterion, uniaxial tensile tests at 0° and 90° , and shear tests at 45° are necessary for all three types. The fourth parameter can be determined from one of three tests: a uniaxial tensile test at 45° (type 1), an equi-biaxial tensile test (type 2), and a pure shear test at 45° (type 3).

For comparison, the yield stresses needed to determine the stress based Hill48 yield criterion [1,19] are also presented in Table 3. The Hill48 yield criterion requires $U_{t,0^\circ}$, $U_{t,45^\circ}$, $U_{t,90^\circ}$, and $B_{t,xy}$ as inputs for material parameter calibration. Fig. 6 provides an overview of the yield surface of H340 at a true plastic strain of 0.002 and a strain rate of 0.001

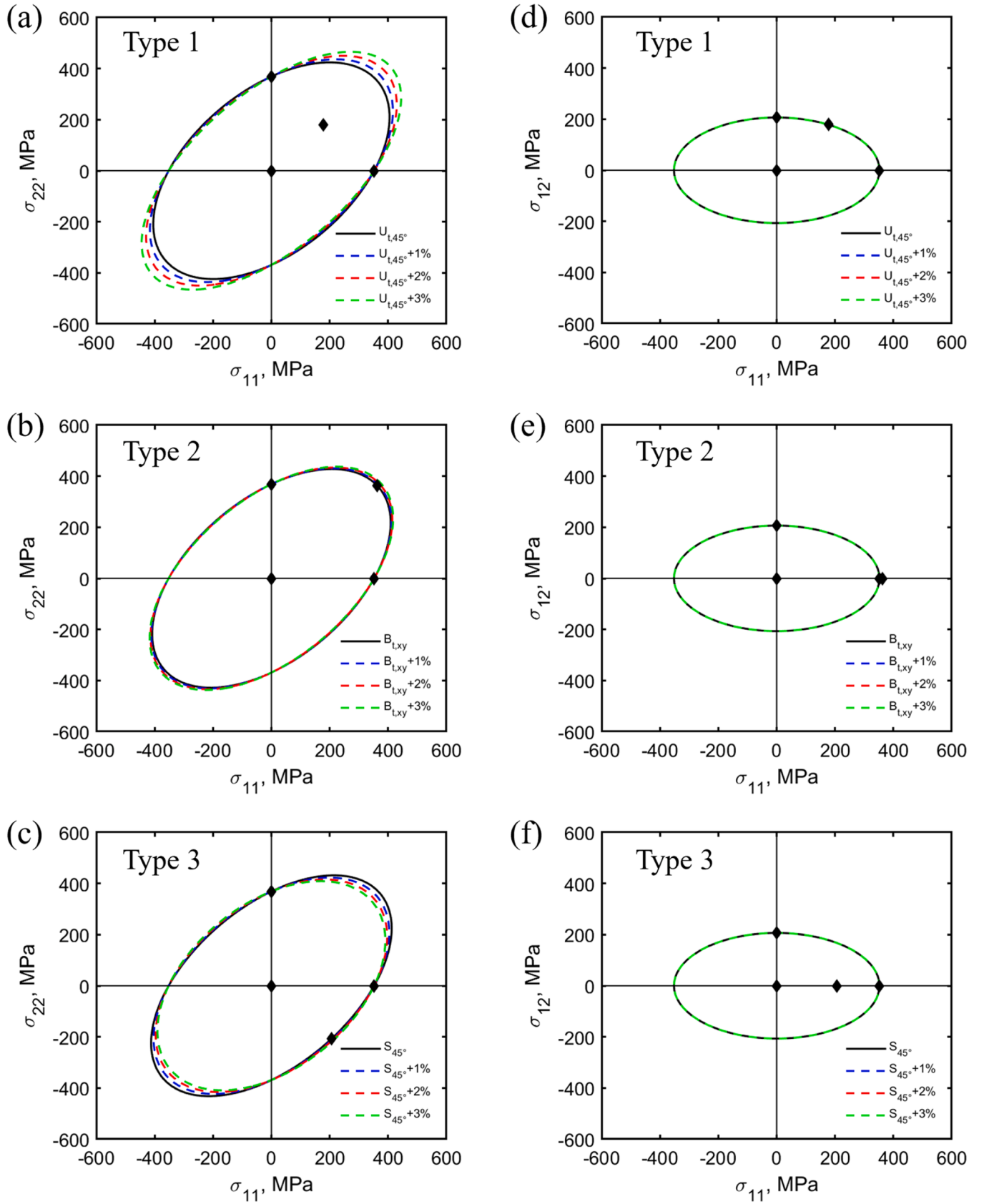


Fig. 7. Prediction of yield loci with three different types of YldSRH yield criteria. (a-c) $\sigma_{12} = 0$, and (d-f) $\sigma_{22} = 0$. The black diamond dots represent the input used to determine the yield surface or loci. The dashed lines show the influences of the measurement deviations of $U_{t,45^\circ}$, $B_{t,xy}$ and S_{45° (ranging from 1% to 3%) on yield loci.

Table 5

Summary of the TCP return mapping algorithm.

1	Compute trial elastic stress $\sigma_{n+1}^{trial} = \sigma^n + \mathbf{C} : \Delta \epsilon_{n+1}$
2	if $[f(\sigma_{n+1}^{trial}) - 1] \cdot \sigma_{ref} \leq 0$ (elastic) then
3	$\Delta \epsilon^p = 0$, $\sigma_{n+1} = \sigma_{n+1}^{trial}$
4	Else
5	for $k = 1$ to N do
6	Calculate the plastic multiplier: $\Delta \lambda^{(k+1)} = \Delta \lambda^{(k)} + \frac{(f^{(k)} - 1) \cdot \sigma_{ref}}{\frac{\delta f^{(k)}}{\delta \sigma} : \mathbf{C} : \frac{\delta f^{(k)}}{\delta \sigma} + H_{(k),0}}$
7	Update: $\sigma_{n+1}^{(k+1)} = \sigma_{n+1}^{(k)} + \mathbf{C} : \left(\Delta \epsilon - \Delta \lambda^{(k+1)} \frac{\delta f^{(k)}}{\delta \sigma} \right)$
8	$\Delta \epsilon_{n+1}^{p(k+1)} = \Delta \epsilon_{n+1}^{p(k)} + \Delta \lambda^{(k+1)}$
9	if $[f(\sigma_{n+1}^{(k+1)}) - 1] \cdot \sigma_{ref} > tol$ (here: $tol = 10^{-9}$ GPa) then
10	$k = k + 1$
11	else
12	Return;
13	end
14	End

s^{-1} , along with some corresponding points under typical stress states. To determine the yield surface in the stress space $\sigma_{11} - \sigma_{22} - \sigma_{12}$, the inclusion of S_{0° in the YldSRH model is helpful. The yield stresses used to determine the yield surface of H340 steel in Fig. 6 are listed in Table 4.

Fig. 7 shows the yield loci with the three types of YldSRH yield criterion in the $\sigma_{11} - \sigma_{22}$ and $\sigma_{11} - \sigma_{12}$ planes. The black diamond dots show the input used to determine the yield loci. The dashed lines show the influences of the measurement deviations of $U_{t,45^\circ}$, $B_{t,xy}$ and S_{45° (ranging from 1% to 3%) in the calculation of the yield loci. It is obvious that the scatter of the 45° tensile stress values (see Fig. 7a) had a more pronounced impact on the yield surface than that of the equi-biaxial tensile (Fig. 7b) or 45° off-axis shear stress values (Fig. 7c). For a deviation of 1% to 3% in the equi-biaxial tensile stress values, the yield loci remained almost the same as the original yield locus. A comparison of the results presented in Fig. 7a-c for type 2 using the equi-biaxial test would provide the most reliable determination of F_{12} and the corresponding yield locus. Therefore, the YldSRH model of type 2 (see Eq. (10)) should be used to characterize the anisotropy of H340 at different strain rates.

In the present study, due to the unavailability of experimental data under dynamic equi-biaxial tension conditions, the flow stress in equi-biaxial tension is estimated by employing an averaged value derived from uniaxial tensile tests conducted in three material orientations: $B_{t,xy} = (U_{t,0^\circ} + 2U_{t,45^\circ} + U_{t,90^\circ})/4$ [20,27]. In this case, uniaxial tensile tests at different loading speeds using SDB and SH specimens (see Fig. 1) were sufficient to determine the anisotropic parameters of the YldSRH model. The hardening curves could be accurately determined using the well-developed experimental techniques of high-speed tensile tests in previous research [55–57,61]. However, it is worth noting that when constructing a yield surface for anisotropic materials, particularly those with tension-compression asymmetry, it is important to consider a wider range of stress states under dynamic loading, including uniaxial compression and equi-biaxial tension and compression [81–88]. Dynamic equi-biaxial tensile testing using servo-hydraulic systems at strain rates above $50 s^{-1}$ is challenging due to force signal oscillations caused by system ringing. Alternatively, Jin et al. [85,87,88] introduced an electromagnetic biaxial split Hopkinson bar system that enables the measurement of material behavior under equi-biaxial loading at very high strain rates ($> 1000 s^{-1}$). Despite this advancement, there is still a clear need for improved experimental setups to perform equi-biaxial tensile tests in the intermediate strain rate range ($50 - 1000 s^{-1}$), as well as dynamic compression tests for automotive thin sheet metals. These works could be performed in the future.

3.1.3. Plastic flow rule

In this study, a tangent cutting plane (TCP) algorithm was applied to return the stress state to the yield surface, which itself expands simultaneously as a result of plastic loading [80,89]. In the TCP approach, the aim is to find a solution, denoted by a Cauchy stress tensor σ and a non-negative scalar factor $\Delta \lambda$, that satisfies the equation $f(\sigma_{n+1}^{trial}) - 1 = 0$ (see Eq. (12)). This requires an iterative process (see Eq. (15) - (16)) to determine the values of σ and $\Delta \lambda$ that satisfy this criterion.

$$\sigma^{(k+1)} = \sigma^{(k)} + \Delta \sigma = \sigma^{(k)} + \mathbf{C} : \left(\Delta \epsilon - \Delta \lambda^{(k+1)} \frac{\delta f}{\delta \sigma} \sigma_{ref} \right). \quad (15)$$

$$\Delta \lambda^{(k+1)} = \Delta \lambda^{(k)} + \Delta(\Delta \lambda). \quad (16)$$

In Eq. (15), k represents the iteration number, $\mathbf{C}$ denotes the elastic stiffness matrix, $\Delta \epsilon$ is the total strain increment, and σ_{ref} is the reference flow stress. The $\Delta(\Delta \lambda)$ in Eq. (16) is the change in the incremental plastic multiplier $\Delta \lambda$. In this study, the $\Delta(\Delta \lambda)$ is defined as:

$$\Delta(\Delta \lambda) = \frac{(f^{(k)} - 1) \cdot \sigma_{ref}}{\frac{\delta f^{(k)}}{\delta \sigma} : \mathbf{C} : \frac{\delta f^{(k)}}{\delta \sigma} + H_{(k),0}}, \quad (17)$$

where $H_{(k),0}$ is the slope of the hardening curve of 0° tensile test at $\bar{\epsilon}_{(k)}^p = \bar{\epsilon}_n^p + \Delta \lambda^{(k)}$ and defined as

$$H_{(k)} = \frac{\delta \sigma_y}{\delta \bar{\epsilon}^p}(\bar{\epsilon}_{(k)}^p). \quad (18)$$

This new form of yield criterion in Eq. (12) has been implemented as a user material subroutine in LS-DYNA (explicit), using the TCP return mapping algorithm. Table 5 shows a short summary of the TCP return mapping algorithm to update the stress and equivalent plastic strain during a time step.

3.2. Rate- and temperature-dependent damage mechanics model

To predict the damage and fracture behavior of sheet metals, many phenomenological damage mechanics models have been developed [47, 61,66,69,90–97]. In this work, the proposed YldSRH model was coupled with a rate- and temperature-dependent damage mechanics model (e^2 MBW) [61], referred to as the YldSRH+ e^2 MBW model. The reason for choosing the e^2 MBW model for damage and fracture modeling of H340 steel is that it is an enhanced damage mechanics model, built upon the MBW and eMBW models [47,66–68]. The e^2 MBW model considers the effects of strain rate and adiabatic temperature on fracture behavior and has been validated in our previous studies [61]. In the following paragraphs, the general formulation regarding damage and fracture in the e^2 MBW model is briefly described.

In the e^2 MBW model, damage is taken into account in the material stress-strain response, and damage evolution is treated macroscopically and phenomenologically. The damage variable D is introduced into the hardening law [66], along with strain rate and temperature dependency [60], as reflected in Eq. (19).

$$\begin{aligned} U_{t,0^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p, T) &= \sigma_{ref,0^\circ}(\bar{\epsilon}^p) f_{t,0^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p) \cdot g(T) \cdot (1 - D); \\ U_{t,45^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p, T) &= \sigma_{ref,45^\circ}(\bar{\epsilon}^p) f_{t,45^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p) \cdot g(T) \cdot (1 - D); \\ U_{t,90^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p, T) &= \sigma_{ref,90^\circ}(\bar{\epsilon}^p) f_{t,90^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p) \cdot g(T) \cdot (1 - D); \\ S_{0^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p, T) &= \sigma_{ref,S,0^\circ}(\bar{\epsilon}^p) f_{s,0^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p) \cdot g(T) \cdot (1 - D). \end{aligned} \quad (19)$$

To characterize the damage variable D , a ductile damage indicator I_{dd} is defined as the integral of the accumulated equivalent plastic strain

Table 6Key equations for the YldSRH+e²MBW plasticity and damage mechanics model.

1. Law of Elasticity
$\boldsymbol{\sigma} = \mathbf{C} : \boldsymbol{\varepsilon}^e$
2. Yield criterion
$f = \left[F_{11}\sigma_{11}^2 + F_{22}\sigma_{22}^2 + \frac{4F_{11}F_{22}}{(\sqrt{F_{11}} + \sqrt{F_{22}})^2}\sigma_{33}^2 + 2F_{12}(\sigma_{11}\sigma_{22} + \sigma_{22}\sigma_{33} + \sigma_{11}\sigma_{33}) + F_{44}(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{13}^2) \right]^{\frac{1}{2}} \leq 1$ $F_{11} = \frac{1}{U_{t,0}(\bar{\varepsilon}^p, \bar{\varepsilon}_p, T)^2}; F_{22} = \frac{1}{U_{t,90}(\bar{\varepsilon}^p, \bar{\varepsilon}_p, T)^2}; F_{44} = \frac{1}{S_0(\bar{\varepsilon}^p, \bar{\varepsilon}_p, T)^2}$ $F_{12} = \frac{1}{2B_{t,xy}^2(\bar{\varepsilon}^p, \bar{\varepsilon}_p, T)} \left[1 - B_{t,xy}^2(\bar{\varepsilon}^p, \bar{\varepsilon}_p, T)(F_{11} + F_{22}) \right] < \pm\sqrt{F_{11}F_{22}}$
3. Associated plastic flow rule
$\Delta\bar{\varepsilon}^p = \Delta\lambda$
$\Delta\lambda^{(k+1)} = \Delta\lambda^{(k)} + \Delta(\Delta\lambda), \Delta(\Delta\lambda) = \frac{(f^{(k)} - 1) \cdot \sigma_{ref}}{\frac{\delta f^{(k)}}{\delta \boldsymbol{\sigma}} : \mathbf{C} : \frac{\delta f^{(k)}}{\delta \boldsymbol{\sigma}} + H_{(k),0}}$
4. Rate- and temperature-dependent hardening law
$U_{t,0}(\bar{\varepsilon}^p, \bar{\varepsilon}_p, T) = \sigma_{ref,0}(\bar{\varepsilon}^p) f_{t,0}(\bar{\varepsilon}^p, \bar{\varepsilon}_p) \cdot g(T) \cdot (1 - D)$ $U_{t,45}(\bar{\varepsilon}^p, \bar{\varepsilon}_p, T) = \sigma_{ref,45}(\bar{\varepsilon}^p) f_{t,45}(\bar{\varepsilon}^p, \bar{\varepsilon}_p) \cdot g(T) \cdot (1 - D)$ $U_{t,90}(\bar{\varepsilon}^p, \bar{\varepsilon}_p, T) = \sigma_{ref,90}(\bar{\varepsilon}^p) f_{t,90}(\bar{\varepsilon}^p, \bar{\varepsilon}_p) \cdot g(T) \cdot (1 - D)$ $S_0(\bar{\varepsilon}^p, \bar{\varepsilon}_p, T) = \sigma_{ref,S,0}(\bar{\varepsilon}^p) f_{s,0}(\bar{\varepsilon}^p, \bar{\varepsilon}_p) \cdot g(T) \cdot (1 - D)$ $B_{t,xy} = (U_{t,0} + 2U_{t,45} + U_{t,90})/4$
5. Damage initiation strain and evolution law
$\bar{\varepsilon}_{ddi}^p(\eta_{avg}, \bar{\theta}_{avg}) = (D_1 e^{-D_2 \eta} - D_3 e^{-D_4 \eta}) \bar{\theta}^2 + D_3 e^{-D_4 \eta}$ $I_{dd} = \int_0^{\bar{\varepsilon}^p} \frac{d\bar{\varepsilon}^p}{\bar{\varepsilon}_{ddi}^p(\eta, \bar{\theta})}$ $D = \int_{\bar{\varepsilon}_{ddi}^c}^{\bar{\varepsilon}^p} m \cdot \left[\frac{\bar{\varepsilon}^p - \bar{\varepsilon}_{ddi}^p}{\bar{\varepsilon}_{df}^p - \bar{\varepsilon}_{ddi}^p} \right]^{m-1} \cdot \frac{d\bar{\varepsilon}^p}{(\bar{\varepsilon}_{df}^p - \bar{\varepsilon}_{ddi}^p)} \cdot D_{cr}, D_{cr} = \frac{\sigma_{ddi}^c}{G_f}$
6. Critical equivalent plastic strain for ductile fracture
$\bar{\varepsilon}_{df}^p(\eta_{avg}, \bar{\theta}_{avg}, \bar{\varepsilon}_p, T) = \left[(F_1^{df} e^{-F_2^{df} \eta} - F_3^{df} e^{-F_4^{df} \eta}) \bar{\theta}^2 + F_3^{df} e^{-F_4^{df} \eta} \right] \cdot \left[1 + \chi(\eta, \bar{\theta}) \cdot \ln\left(\frac{\bar{\varepsilon}_p}{\varepsilon_0}\right) \right] \cdot \left(1 + F_5^{df} \cdot \frac{T - T_0}{T_m - T_0} \right)$

[47,66]:

$$I_{dd} = \int_0^{\bar{\varepsilon}^p} \frac{d\bar{\varepsilon}^p}{\bar{\varepsilon}_{ddi}^p(\eta, \bar{\theta})}, \quad (20)$$

where $\bar{\varepsilon}_{ddi}^p$ is a ductile damage initiation (DDI) criterion [66] that is stress-state dependent and defined as follows:

$$\bar{\varepsilon}_{ddi}^p(\eta_{avg}, \bar{\theta}_{avg}) = [(D_1 e^{-D_2 \eta} - D_3 e^{-D_4 \eta}) \bar{\theta}^2 + D_3 e^{-D_4 \eta}], \quad (21)$$

where $D_i = 1 - 4$ are material constants, η is the stress triaxiality, and $\bar{\theta}$ is the Lode angle parameter.

When the ductile damage indicator I_{dd} is less than 1, the plastic strain accumulates without any damage. Once I_{dd} reaches 1, damage begins to occur, and the subsequent ductile damage evolution follows an energy-based law, as described in Eq. (22) [61]:

$$\bar{\varepsilon}_{df}^p(\eta_{avg}, \bar{\theta}_{avg}, \bar{\varepsilon}_p, T) = \left[(F_1^{df} e^{-F_2^{df} \eta} - F_3^{df} e^{-F_4^{df} \eta}) \bar{\theta}^2 + F_3^{df} e^{-F_4^{df} \eta} \right] \cdot \left[1 + \chi(\eta, \bar{\theta}) \cdot \ln\left(\frac{\bar{\varepsilon}_p}{\varepsilon_0}\right) \right] \cdot \left(1 + F_5^{df} \cdot \frac{T - T_0}{T_m - T_0} \right), \quad (24)$$

$$D = \int_{\bar{\varepsilon}_{ddi}^c}^{\bar{\varepsilon}^p} m \cdot \left[\frac{\bar{\varepsilon}^p - \bar{\varepsilon}_{ddi}^p}{\bar{\varepsilon}_{df}^p - \bar{\varepsilon}_{ddi}^p} \right]^{m-1} \cdot \frac{d\bar{\varepsilon}^p}{(\bar{\varepsilon}_{df}^p - \bar{\varepsilon}_{ddi}^p)} \cdot D_{cr}. \quad (22)$$

The parameter m in Eq. (22) is a material constant that must be either

calibrated through FE simulations or determined by experiments. It dictates the damage accumulation rate after DDI, where $m=0$ implies no damage (uncoupled model) [47,57,74], $m=1$ indicates linear damage evolution [66,69,98], and $m>1$ suggests a nonlinear relationship with plastic strain [90,92,99,100]. $\bar{\varepsilon}_{df}^p$ is the critical equivalent plastic strain for ductile fracture (DF). D_{cr} is a critical damage value which is defined as follows:

$$D_{cr} = \frac{\sigma_{ddi}^c}{G_f}, \quad (23)$$

where the variable σ_{ddi}^c represents the equivalent stress at the damage initiation point, and G_f denotes the fracture energy of the material. Fracture occurs when the damage D reaches the critical value D_{cr} .

The $\bar{\varepsilon}_{df}^p$ in Eq. (22) is stress-state, strain-rate and temperature dependent and is defined as follows [61]:

where $F_{i=1-5}^{df}$ are material constants, and T_0 and T_m are the room and melting temperature, respectively. $\chi(\eta, \bar{\theta})$ is a stress state-dependent term used to describe the complex effects of strain rate on fracture strain [61].

For a clear and comprehensive overview of YldSRH+e²MBW

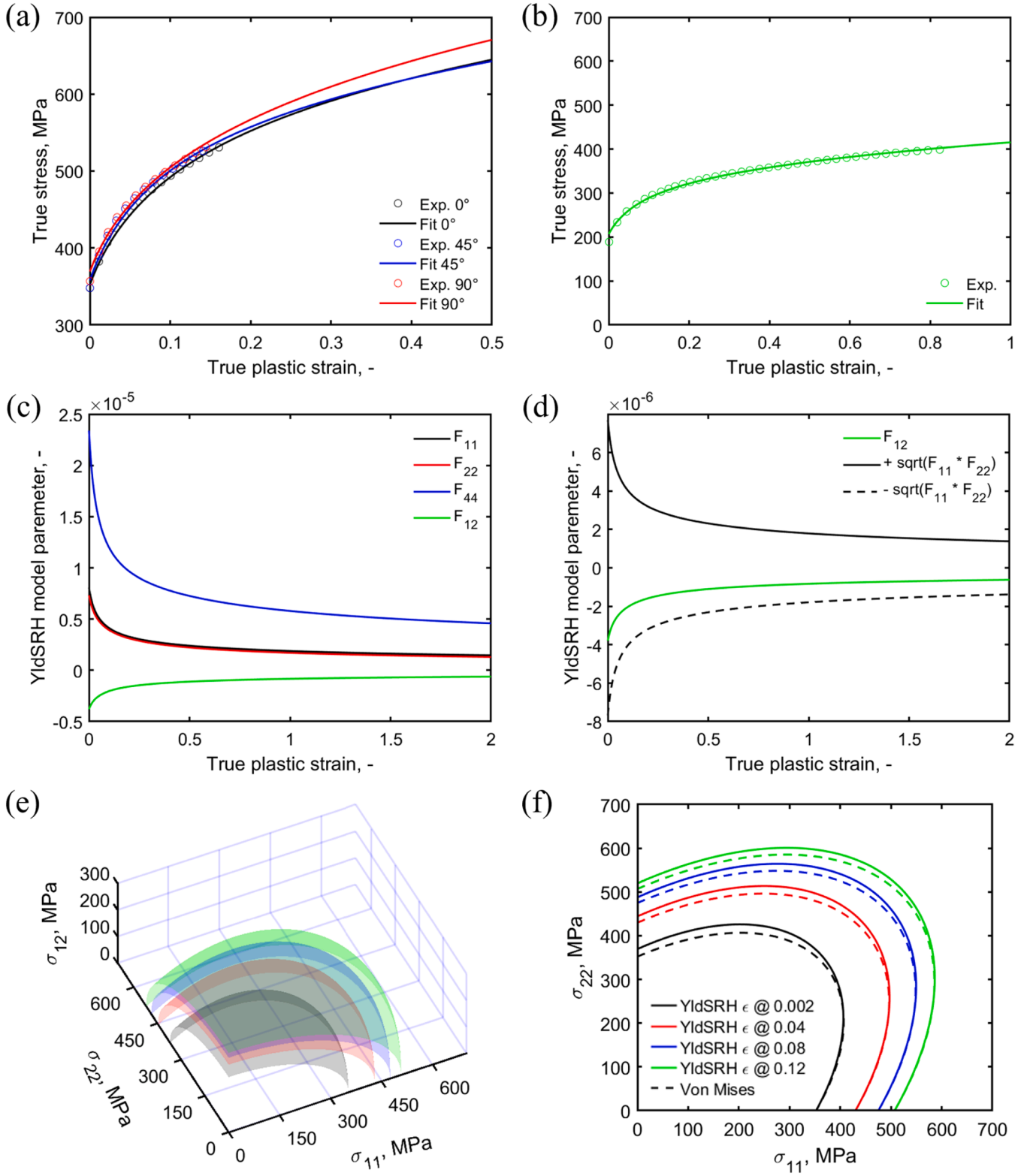


Fig. 8. Experimental and fitted flow curves, along with the YldSRH model parameters and the corresponding yield surface and loci. (a-b) The flow curves of uniaxial tension at 0°, 45°, and 90° in the RD and 0° SH test under QS condition. (c) Evolutions of model parameters F_{11} , F_{22} , F_{44} , and F_{12} with true plastic strain. (d) Comparison of the values of F_{12} and $\pm \sqrt{F_{11}F_{22}}$ at different true plastic strains. (e) Yield surface (1/8) at different strains (0.002, 0.04, 0.08 and 0.12) and a strain rate of 0.001 s⁻¹ (f) Evolution of yield loci with strain in the σ_{11} - σ_{22} plane.

Table 7

Calibrated parameters for the plasticity of H340 using combined Swift-Voce law.

	A	$\bar{\epsilon}_0$	n	k_0	Q	β	α
$\sigma_{ref,0^\circ}$	755	0.024	0.206	359	195	12.3	0.85
$\sigma_{ref,45^\circ}$	760	0.023	0.200	361	183	15.1	0.80
$\sigma_{ref,90^\circ}$	762	0.026	0.199	362	183	15.1	1
$\sigma_{ref,S,0^\circ}$	439	0.023	0.200	208	105	15.1	0.80

plasticity and damage mechanics model, the most important equations are summarized in Table 6.

4. Model parameter calibration and validation

The YldSRH+e²MBW model, as proposed in Section 3, was applied to H340 steel sheets under the various loading conditions described in Section 2. The subsequent sections detail the step-by-step calibration and validation of the material parameters for the YldSRH+e²MBW model. Section 4.1 presents the calibration and identification of the anisotropy parameters of the YldSRH model, accounting for strain rate and temperature effects. Section 4.2 discusses the calibration of damage and fracture-related parameters. Finally, Section 4.3 validates the YldSRH+e²MBW model by comparing the simulation results with experimental data.

4.1. Anisotropic plasticity parameter calibration and identification

Unlike the traditional dynamic plasticity models that typically consider strain rate and temperature dependence only for the 0° direction [44,46,47,61,74], the YldSRH model incorporates four sets of independent input parameters, as given in Eq. (19). The reference flow

curves $\sigma_{ref}(\bar{\epsilon}^p)$ and strain rate sensitivity term $f(\bar{\epsilon}^p, \bar{\epsilon}_p)$ were calibrated using, for example, 0°, 45°, and 90° tensile test, respectively. Other effects such as the temperature effect $g(T)$ and damage softening $(1 - D)$ were calibrated based on the test results in the RD.

4.1.1. Parameter calibration under RT and QS conditions

In this work, a combined Swift-Voce law (see Eq. (25)) was employed to obtain the reference flow stress $\sigma_{ref}(\bar{\epsilon}^p)$ by extrapolating the experimental true stress-true plastic strain curve. Repeated simulations were conducted to refine the flow curve by adjusting the value of α in Eq. (25) within the post-necking regime. This process continued until the force-displacement curves from the simulations for different specimen geometries closely align with the experimental results.

$$\sigma_{ref}(\bar{\epsilon}^p) = \bar{\sigma}_y^{S-V}(\bar{\epsilon}^p) = \alpha A (\bar{\epsilon}_0 + \bar{\epsilon}^p)^n + (1 - \alpha) \cdot [k_0 + Q(1 - e^{-\beta \bar{\epsilon}^p})]. \quad (25)$$

Fig. 8a-b shows a comparison between the experimental and fitting flow curves of H340 for uniaxial tension at 0°, 45°, and 90° and the SH

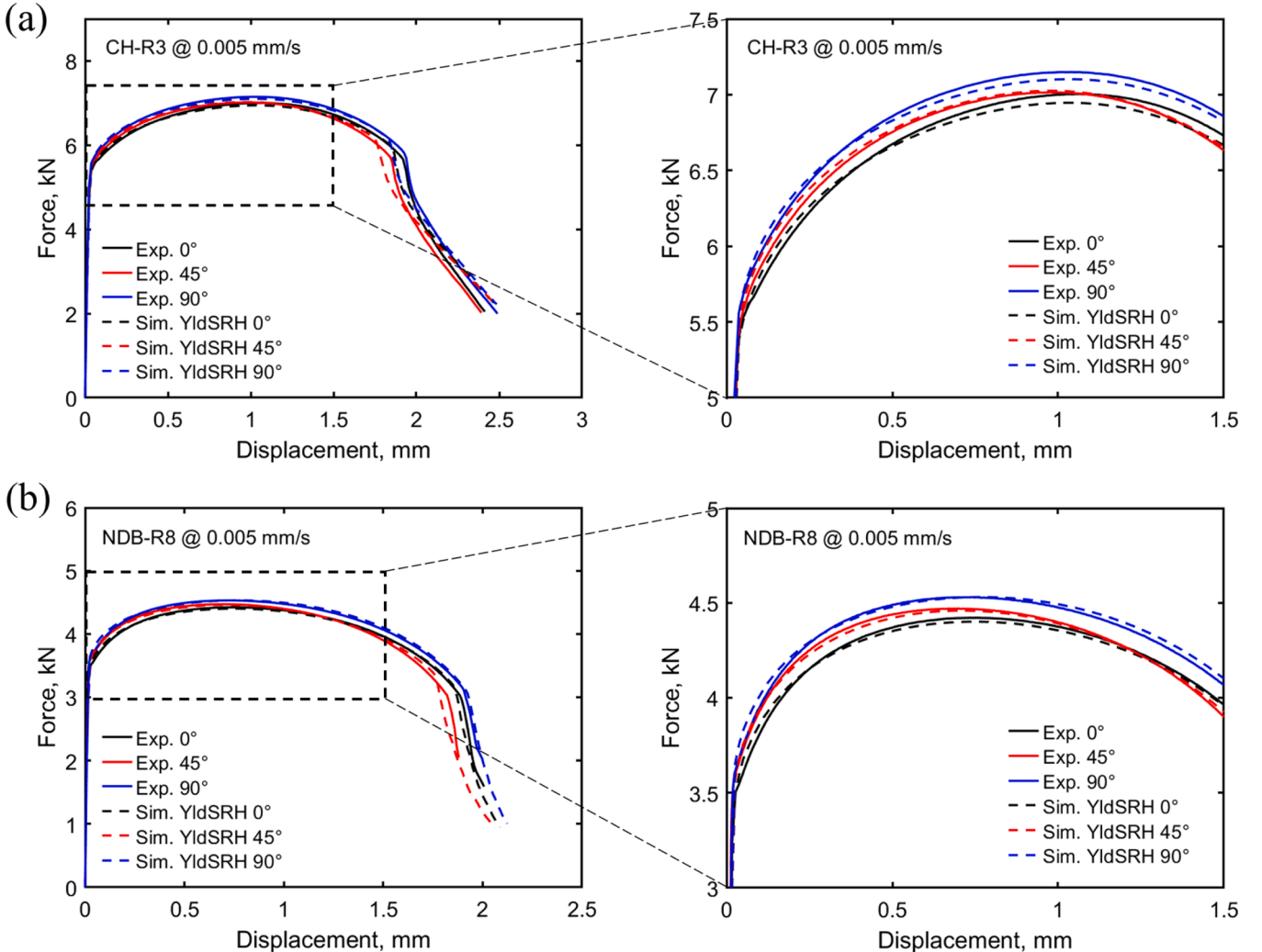


Fig. 9. Experimental and numerical force-displacement curves of the tensile test results of the (a) CH-R3 and (b) NDB-R8 specimens in the 0°, 45°, and 90° directions.

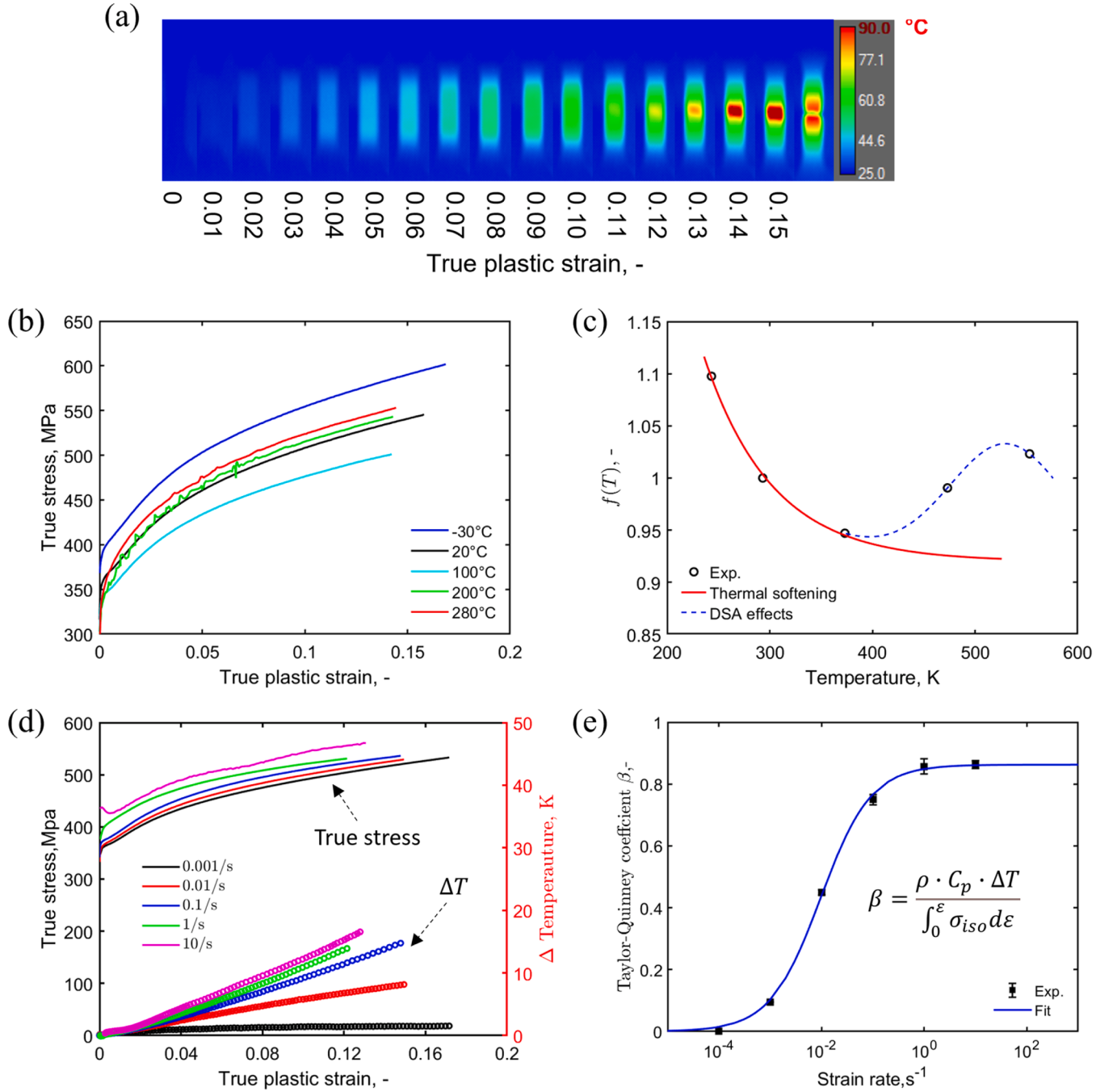


Fig. 10. Thermal effect parameter determination and fitting. (a) Temperature field of the gauge section of SDB specimen during a uniaxial tensile test at a strain rate of 1 s^{-1} . (b) True stress-true plastic strain curves at different temperatures (-30 - 280°C) under QS conditions (0.001 s^{-1}). (c) Experimentally measured temperature dependence of flow stress $g(T) = \sigma_T/\sigma_{RT}$ at $\bar{\varepsilon}^p = 0.02$ and its curve fitting using an exponential function. The dash line shows the DSA effect observed at QS condition. (d) True stress-true plastic strain curves and the corresponding temperature increase versus true plastic strain curves at strain rate range of $0.001 - 10 \text{ s}^{-1}$. (e) Taylor-Quinney coefficient β as a function of strain rate. The β at 0.0001 s^{-1} was set as 0, as almost no temperature increase was observed at this strain rate.

Table 8

Calibrated parameters for the thermal effects of H340.

C_1^T	C_2^T	C_3^T	$\rho, \text{kg/m}^3$	$C_p, \text{J/(kg} \cdot \text{K)}$	β_i	S_c	$\bar{\varepsilon}_c$
6.767	0.015	0.920	7870	466	0.87	0.44	0.0098

test in the 0° direction. The fitted flow curve will be used as a reference for the calibration of the temperature and strain-rate sensitivity parameters. The calibrated parameters A , $\bar{\varepsilon}_0$, n , k_0 , Q , β , and α are listed in Table 7.

With the fitted hardening curves shown in Fig. 8a-b, the value of F_{11} , F_{22} , F_{44} and F_{12} were calculated, as shown in Fig. 8c. It should be noted that, the type 2 YldSRH model (see Fig. 7) was adopted here, as the equibiaxial tensile stress was calculated as an average value using $B_{t,xy} = (U_{t,0^\circ} + 2U_{t,45^\circ} + U_{t,90^\circ})/4$. However, this assumption may lead to inaccuracies in predicting plasticity under biaxial stress states. For

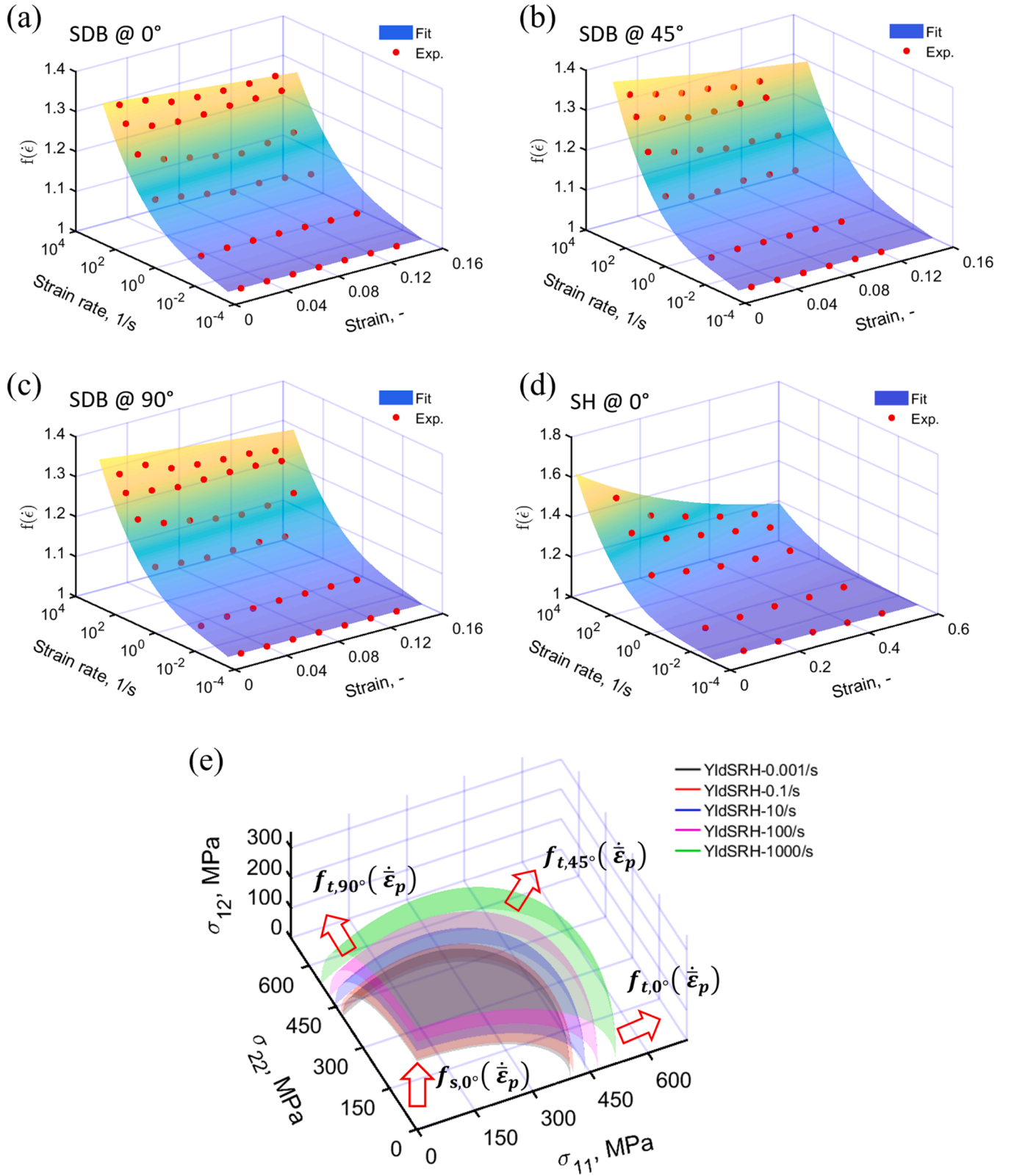


Fig. 11. Strain rate sensitivity parameter fitting and YldSRH yield surface of H340. (a-d) Experimental (red dot) and fitted (surface) strain rate dependence of SDB tensile tests in the 0°, 45°, and 90° loading directions, and 0° SH test. (e) YldSRH Yield surface (1/8) of H340 at a strain of 0.02 and different strain rates (0.001, 0.1, 10, 100, and 1000 s^{-1}). The arrows indicate that the yield surface expands non-uniformly with increasing strain rate.

future work, conducting high-speed biaxial tests would provide more precise data.

Fig. 8d shows the evolution of F_{12} , calculated using Eq. (10), and $\pm \sqrt{F_{11}F_{22}}$ with the true plastic strain. It can be observed that with the given input data, the inequality condition in Eq. (10) is always satisfied, ensuring the formation of a closed yield surface. Fig. 8e shows the one-eighth (1/8) yield surface at different plastic strains. The corresponding yield loci at different strains are plotted in the $\sigma_{11} - \sigma_{22}$ plane in Fig. 8f. The dashed lines show the yield loci based on the von Mises criterion, using only the flow curves of the uniaxial tension at 0°.

To validate the predictive capability of the YldSRH model for yielding and anisotropic hardening of H340, FE simulations were conducted using the model coupled with the damage mechanics model (e²MBW). These simulations were performed in an LS-DYNA explicit solver under near-QS conditions (0.005 mm/s). Solid elements of size $0.1 \times 0.1 \times 0.1 \text{ mm}^3$ were used for the major deformation zone (gauge section) of specimens. To reduce simulation time for a full model with a solid mesh size of 0.1 mm and the symmetry of the mechanical system, only a quarter of each specimen within the gauge section was modeled. Fig. 9 shows the experimental and numerical force-displacement curves in the tensile tests for the CH-R3 and NDB-R8 specimens in the 0°, 45°, and 90° directions. The results presented in Fig. 9 show that the predictive ability of the YldSRH model, calibrated using data from the SDB and SH specimens (see Fig. 8), had good agreement with the experimental results in the force-displacement curves for the three loading directions of the CH-R3 and NDB-R8 specimens.

4.1.2. Thermal effect parameter calibration

To model the mechanical properties of H340 under a high strain rate, the effects of temperature resulting from adiabatic heating must be considered [46,47,101]. An example of the measured temperature field during a uniaxial tensile test at a strain rate of 1 s^{-1} , captured by a thermal camera, is shown in Fig. 10a. To investigate the thermal effects, uniaxial tensile tests were performed under QS conditions (0.001 s^{-1}) at different temperatures (-30°C, 20°C, 100°C, 200°C and 280°C), since the maximum local temperature near failure of different tensile specimens at the highest loading speed (10000 mm/s) can exceed 200°C, as detailed in our previous study [61]. The true stress-true plastic strain curves measured under QS conditions (see Fig. 10b) indicate that the temperature has a softening effect on material strength for $T \leq 100^\circ\text{C}$, followed by a dynamic strain aging (DSA) effect for $T > 100^\circ\text{C}$, leading to an increase in material strength. However, as noted in the studies by Li et al. [32] and Chandran et al. [102,103], DSA effect (see Fig. 10c)

was deactivated at high strain rates ($\bar{\epsilon}_p > 1 \text{ s}^{-1}$), with temperature sensitivity showing a monotonically decreasing trend as the temperature increases. In addition, as shown in Figs. 3 and 4, the force-displacement curves under high-speed loading conditions do not exhibit sudden strength jumps or increase near material failure period, where the local temperature exceeds 100°C. Therefore, in this work, the temperature sensitivity $g(T)$ was fitted using an exponential function based only on experimental results at -30°C, 20°C and 100°C, as shown in Fig. 10c.

The experimental $g(T)$ shown in Fig. 10c were calculated at a true plastic strain of 0.02 by comparing the true stress at various

temperatures with that at 20°C (i.e., $g(T) = \sigma_T / \sigma_{RT}$). Then the $g(T)$ values were fitted using an exponential function, as shown in Eq. (26). The fitted parameters for the temperature term are listed in Table 8.

$$g(T) = C_1^T \cdot \exp(-C_2^T \cdot T) + C_3^T. \quad (26)$$

Then, the temperature change due to adiabatic heating can be implemented in the user-defined material model and considered in FE simulations. This was treated as an internal state variable, determined using the following equation:

$$\Delta T = \frac{\beta \cdot \int_0^{\epsilon} \sigma_{iso} d\epsilon}{\rho \cdot C_p}, \quad (27)$$

where ΔT is the temperature increase in the specimen during plastic deformation. The parameters ρ and C_p represent the material density and heat capacity, respectively. The fraction β refers to the Taylor-Quinney coefficient [104], which was calculated based on the true stress-true plastic strain curves and the temperature increase versus true plastic strain curves under different strain rates (see Fig. 10d). The experimentally measured β of H340, as shown in Fig. 10e, was determined at an equivalent plastic strain of 0.12 for all different strain rates using a MATLAB tool. A hyperbolic tangent function, as shown in Eq. (28), was used for the fitting of β . The fitted parameters for this function are listed in Table 8.

$$\beta = \frac{\beta_i}{2} \left\{ 1 + \tanh \left[S_c \cdot \log \frac{\bar{\epsilon}_p}{\bar{\epsilon}_c} \right] \right\}. \quad (28)$$

4.1.3. Strain rate sensitivity parameter calibration

After obtaining the temperature effects $g(T)$ and the value of parameter β in Section 4.1.2, the isothermal true stress-true plastic strain curves $\sigma_{iso}(\bar{\epsilon}^p)$ of H340 was calculated from the experimental true stress-true plastic strain curves $\sigma_{adi}(\bar{\epsilon}^p)$ (see Fig. 2a-d) based on the Eq. (29):

$$\sigma_{iso}(\bar{\epsilon}^p) = \frac{\sigma_{adi}(\bar{\epsilon}^p)}{g(T)}. \quad (29)$$

The calculated isothermal true stress-true plastic strain curves were used to determine the strain rate sensitivities $f(\bar{\epsilon}^p, \bar{\epsilon}_p)$ in Eq. (19) for the 0°, 45°, 90° tensile and 0° shear tests. The results are shown in Fig. 11a-d, where the red dots represent the experimental strain rate dependence, calculated based on $f(\bar{\epsilon}_p) = \sigma_{\bar{\epsilon}_p} / \sigma_{0.001 \text{ s}^{-1}}$ and the surfaces illustrate the fitted strain rate dependence using a modified Johnson-Cook (J-C) strain rate sensitivity term given in Eq. (30) [61]. The parameters $C_2^{\bar{\epsilon}}, C_3^{\bar{\epsilon}}$ and $C_4^{\bar{\epsilon}}$ define a nonlinear strain rate sensitivity and its dependence on strain hardening. As illustrated in Fig. 2e-f, the experimentally determined strain rate sensitivity $f(\bar{\epsilon}_p)$ for H340 exhibits a nonlinear relationship with the logarithmic term $\ln \left(\frac{\bar{\epsilon}_p}{\bar{\epsilon}_0} \right)$. Moreover, this nonlinear relationship

is strongly dependent on the loading direction. When $C_4^{\bar{\epsilon}} = 0$, strain rate sensitivity is strain hardening independent. When $C_2^{\bar{\epsilon}} = C_3^{\bar{\epsilon}} = C_4^{\bar{\epsilon}} = 0$, Eq. (30) is simplified to J-C linear strain rate sensitivity term, expressed as

$f(\bar{\epsilon}_p) = 1 + C_1 \left(\frac{\bar{\epsilon}_p}{\bar{\epsilon}_0} \right)^{\bar{m}}$, which is widely used in the literature to describe the strain rate sensitivity of metals [44,46,47,88,105]. The fitted parameters for the four tests conducted on the H340 steel are listed in Table 9.

Table 9

Parameters for fitting strain rate sensitivity under different loading conditions.

	$C_1^{\bar{\epsilon}}$	$C_2^{\bar{\epsilon}}$	$C_3^{\bar{\epsilon}}$	$C_4^{\bar{\epsilon}}$	$\bar{\epsilon}_0, \text{ s}^{-1}$
$f_{t,0^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p)$	0.0040	-0.00035	0.00012	1.0	0.001
$f_{t,45^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p)$	0.0048	-0.00032	0.00013	1.2	
$f_{t,90^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p)$	0.0037	-0.00010	0.00010	1.0	
$f_{s,0^\circ}(\bar{\epsilon}^p, \bar{\epsilon}_p)$	0.0045	-0.00020	0.00020	1.8	

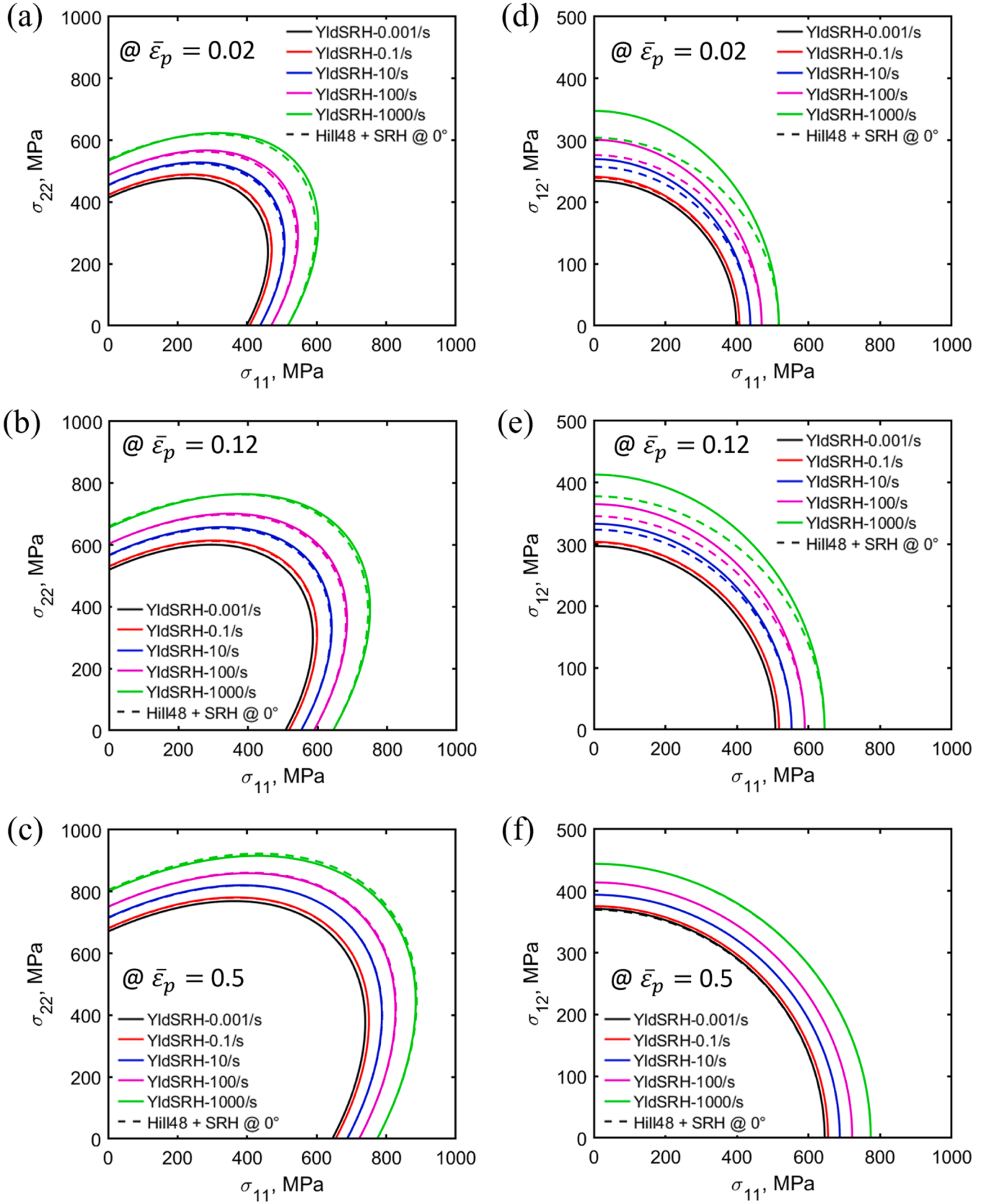


Fig. 12. Evolution of the yield loci of YldSRH (solid line) with strain rates at different strains in the $\sigma_{11} - \sigma_{22}$ and $\sigma_{11} - \sigma_{12}$ planes. (a-b) $\bar{\epsilon}^p = 0.02$, (c-d) $\bar{\epsilon}^p = 0.12$, (e-f) $\bar{\epsilon}^p = 0.5$. The dashed lines show the Hill48 yield loci with uniform strain rate sensitivity determined using a uniaxial tension test in the 0° direction.

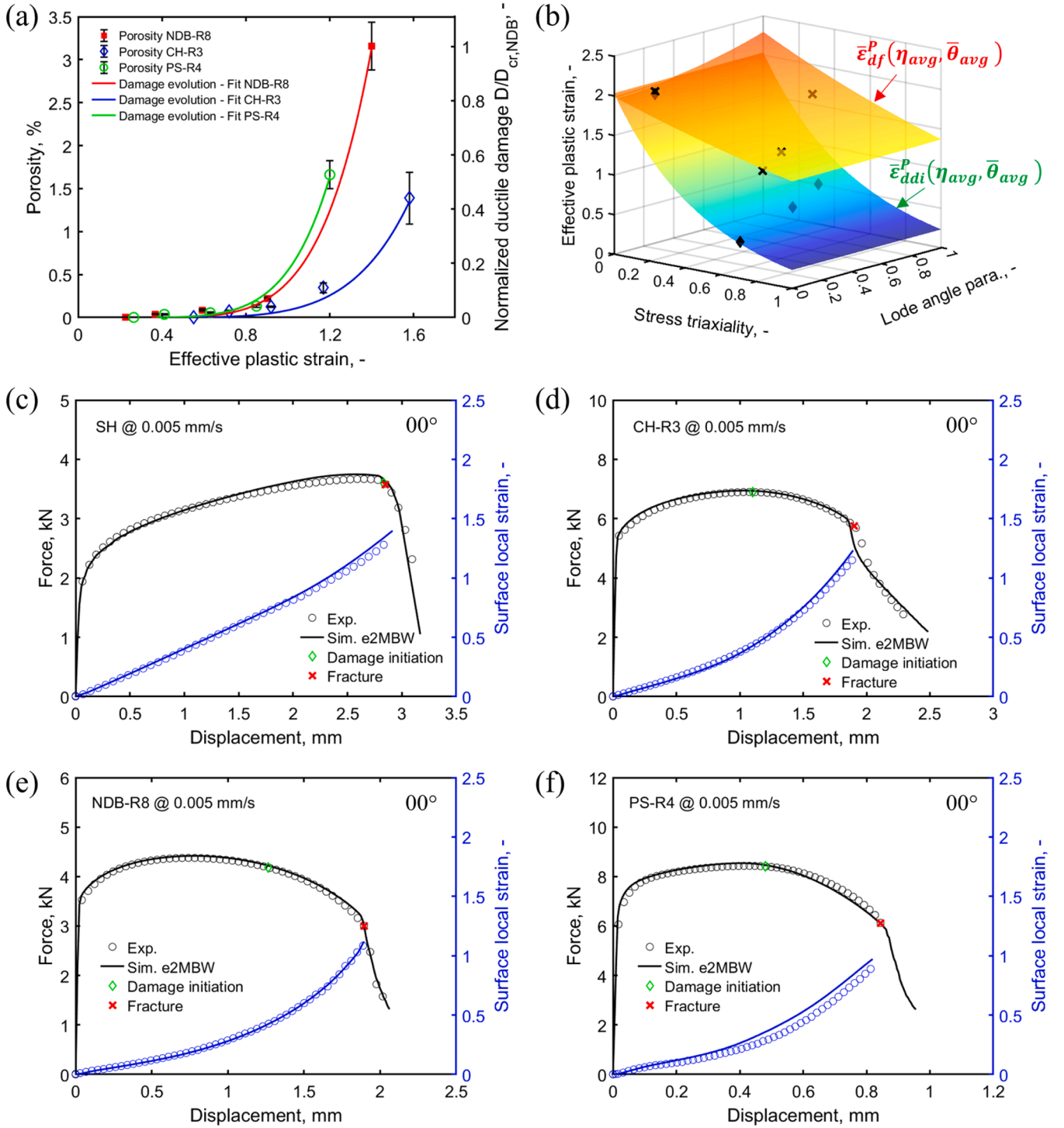


Fig. 13. Calibration of damage- and fracture-related parameters. (a) Porosity evolution of three geometries at various effective plastic strain values, along with normalized ductile damage based on $D_{cr,NDB}$, fitted using $m = 6$ in Eq. (22). (b) Calibration and fitting of the ductile damage initiation (DDI) and ductile fracture (DF) loci for H340 under QS loading conditions. (c-f) Comparison of experimental and simulated (e²MBW) force-displacement curves and surface local strain versus displacement curves for four different specimen geometries, tested along the RD at a loading speed of 0.005 mm/s.

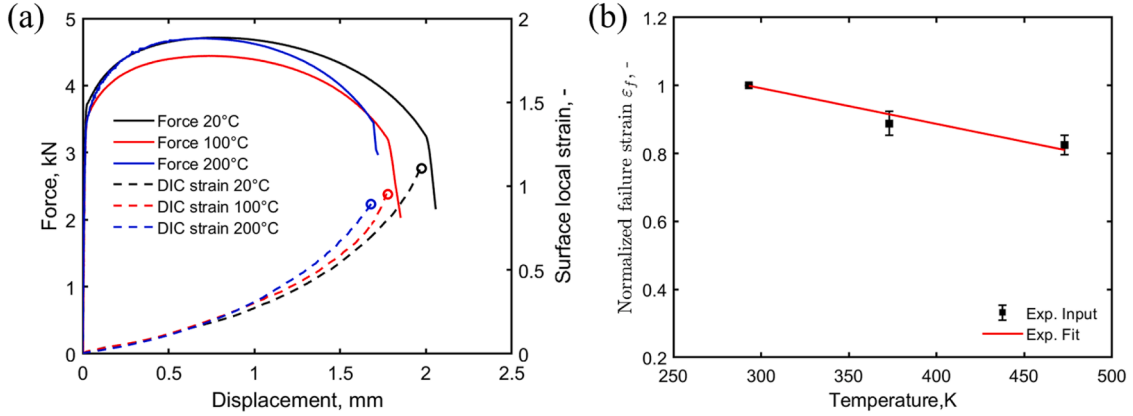
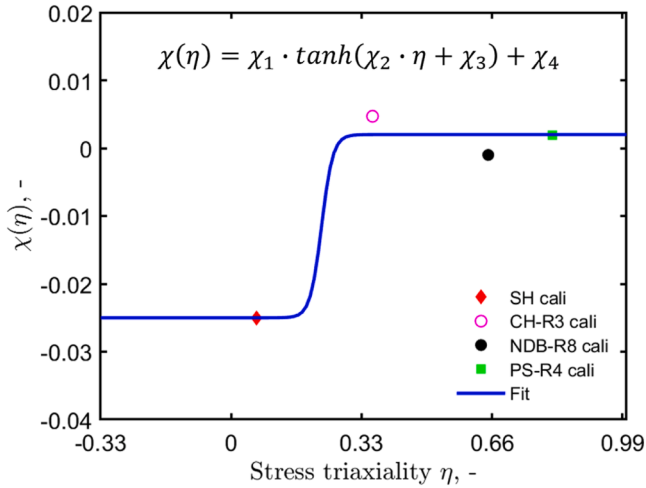
$$f(\bar{\epsilon}^p, \dot{\bar{\epsilon}}^p) = 1 + \left\{ C_1 \ln\left(\frac{\dot{\bar{\epsilon}}^p}{\dot{\bar{\epsilon}}_0}\right) + C_2 \left[\ln\left(\frac{\dot{\bar{\epsilon}}^p}{\dot{\bar{\epsilon}}_0}\right) \right]^2 + C_3 \left[\ln\left(\frac{\dot{\bar{\epsilon}}^p}{\dot{\bar{\epsilon}}_0}\right) \right]^3 \right\} \cdot e^{-C_4 \bar{\epsilon}^p}. \quad (30)$$

Fig. 11a-d shows that the strain rate sensitivities cover a wide range of strain rates, from 0.001 s^{-1} to more than 1000 s^{-1} , and have no discontinuities. Moreover, it is evident that strain rate sensitivity is strain

dependent and particularly stress-state dependent. For uniaxial tension in the 0° direction at a strain rate of 1000 s^{-1} , the strain rate sensitivity decreased from 1.32 to 1.28 as the strain increased from 0 to 0.12. However, for the 0° shear test, the strain rate sensitivity was initially 1.64 but then decreased significantly to 1.2 when the strain reached 0.5. The notable differences in strain rate sensitivity across different loading cases and stress states highlight the necessity of considering the non-

Table 10Calibrated parameters for the damage and fracture of H340 in the e²MBW model.

Damage-related parameters								
m	$G_{f,NDB}$, GPa	$G_{f,CH}$, GPa	$G_{f,PS}$, GPa	D_1	D_2	D_3	D_4	
6	2.0	4.7	3.6	2.034	2.179	2.034	2.179	
Fracture-related parameters								
F_1^{df}	F_2^{df}	F_3^{df}	F_4^{df}	F_5^{df}	χ_1	χ_2	χ_3	χ_4
2.307	0.521	1.997	0.450	-1.110	0.0135	35	-8	-0.0115

**Fig. 14.** Temperature effects on the fracture strain and its curve fitting. (a) Force-displacement and surface local strains-displacement curves at 0.005 mm/s and different temperatures, (b) Fitting of the effects of temperature on fracture strain using a J-C temperature-dependent term.**Fig. 15.** Calibrated value of $\chi(\eta)$ in Eq. (24) and its fitting using a hyperbolic tangent function.

uniform effects of strain rate on anisotropic yielding and hardening under very high strain rate loading conditions.

On the basis of the reference flow stress $\sigma_{ref}(\bar{\epsilon}^p)$ shown in Fig. 8a-b and the strain rate sensitivity presented in Fig. 11a-d, yield surfaces at different strain and strain rates can be obtained using the proposed YldSRH model. Fig. 11e shows the yield surfaces at a strain of 0.02 under different strain rates (0.001, 0.1, 10, 100 and 1000 s⁻¹). The strain rate hardening in the various directions of the stress-space is different, causing a non-uniform expansion of the yield surface with strain at different strain rates, which can be seen more clearly in Fig. 12.

The evolution of the corresponding yield loci with strain rates at different strain values (0.02, 0.12, and 0.5) and with $\sigma_{12} = 0$ and $\sigma_{22} = 0$ is presented in Fig. 12. The dashed lines in Fig. 12 represent the Hill48 yield loci, calculated using the uniform strain rate sensitivity derived

solely from 0° uniaxial tension. The Hill48 yield criterion and the method for calibrating the corresponding parameters are detailed in Appendix C. For lower strains ($\bar{\epsilon}^p = 0.02$ and $\bar{\epsilon}^p = 0.12$), the Hill48 yield loci were almost identical to the YldSRH yield loci at low strain rates (e. g., 0.001 and 0.1 s⁻¹). However, at higher strain rates (10, 100, and 1000 s⁻¹), significant differences were observed between the Hill48 and YldSRH models. The effects of strain rate hardening cannot be accurately described by 0° tensile tests alone, a fact that has been mostly ignored so far.

However, when the strain increased from 0.12 to 0.5, the strain rate sensitivity at the shear stress state decreased to a value close to that at 0° uniaxial tension, as can be seen in Fig. 12d-f. Moreover, the effects of non-uniform strain rate hardening on the yield surface in the $\sigma_{11} - \sigma_{22}$ plane are smaller than those in the $\sigma_{11} - \sigma_{12}$ plane. What should be mentioned is that the equi-biaxial stress values used for the YldSRH model were calculated as an average of the 0°, 45°, and 90° uniaxial tensile stresses, which may underestimate the strain rate sensitivity in equi-biaxial stress states. Further investigations are currently underway, and their results will be published in the future.

4.2. Calibration of damage- and fracture-related parameters

To fully describe the material behavior, the anisotropic plasticity described in Section 4.1 must be coupled with damage and fracture properties. In this work, damage- and fracture-related parameters were determined using specimens oriented in the RD. First, the damage parameter D in Eq. (19) was determined using porosity measurements on scanning electron micrographs (SEM) in combination with FE simulation. In Fig. 13a, the porosity in the center of the gauge section of the NDB-R8, CH-R3, and PS-R4 specimens at various effective plastic strains is presented, along with the approximated normalized damage curve according to Eq. (22), with $m = 6$. The value of parameter m was determined by fitting of the porosity measured at the center of the gauge section of different specimens with the corresponding effective plastic strain at the same position. The effective plastic strain depicted in Fig. 13a was determined from FE simulations, as the DIC only measured

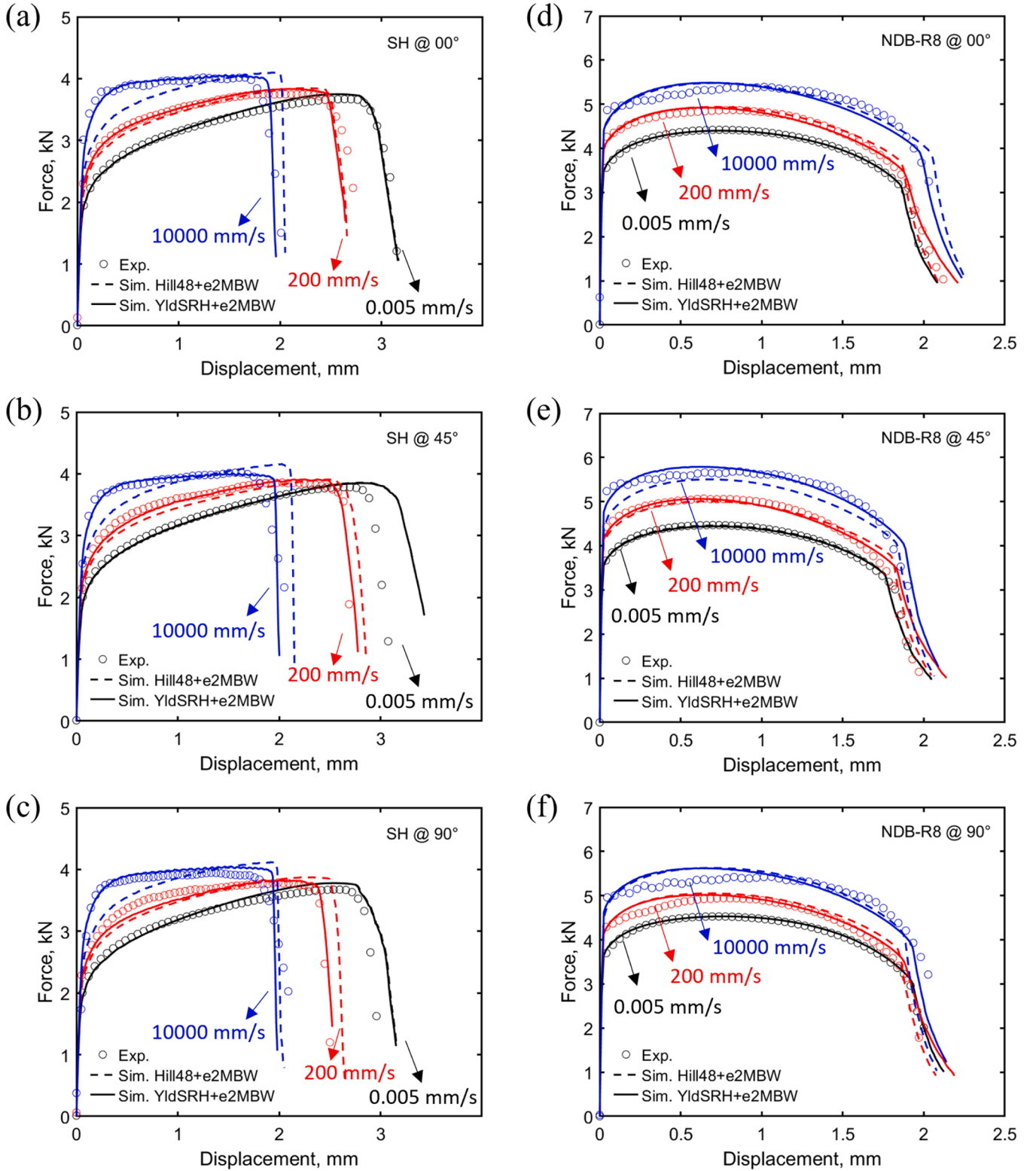


Fig. 16. Experimental and numerical force-displacement curves of H340 at low, intermediate and high loading speeds. (a-c) SH and (d-f) NDB-R8. The solid lines represent the simulation results using the YldSRH+e²MBW model, while the dashed lines correspond to the Hill48+e²MBW models. The Hill48+e²MBW model considered strain rate sensitivity only at 0°.

surface strain and the damage initiated from the center of the specimens. To calibrate the DDI, SEM micrographs were taken along the tensile direction at intervals of approximately 56 μm from the fracture edge to calibrate ductile damage initiation, with corresponding porosities measured. These porosities were then associated with effective plastic

strain values, with the damage initiation threshold defined as a porosity below 0.05%. The calibrated DDI values and its fitting using Eq. (21) are shown in Fig. 13b. The values of damage parameter G_f in Eq. (23) were calibrated through a combination of experimental tests and FE simulations. In addition, the ductile fracture strain under QS conditions, used

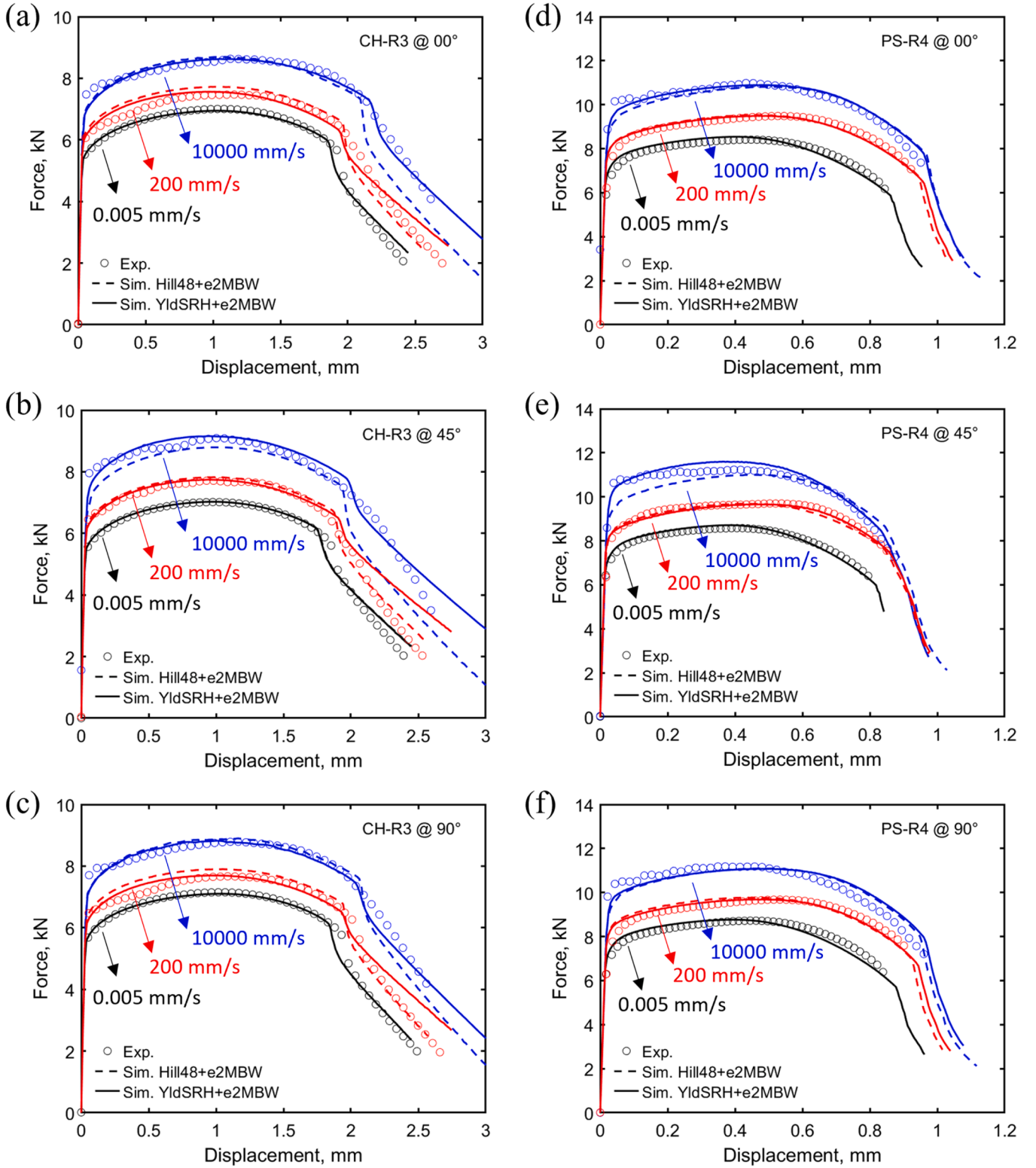


Fig. 17. Experimental and numerical force-displacement curves of H340 at low, intermediate and high loading speeds. (a-c) CH-R3 and (d-f) PS-R4. The solid lines represent the simulation results using the YldSRH+e²MBW model, while the dashed lines correspond to the Hill48+e²MBW models. The Hill48+e²MBW model considered strain rate sensitivity only at 0°.

for fitting the DF locus in Eq. (24), expressed as $\bar{\sigma}_{df}^p = (F_1^{df} e^{-F_2^{df} \eta} - F_3^{df} e^{-F_4^{df} \eta}) \bar{\sigma}^2 + F_3^{df} e^{-F_4^{df} \eta}$, was calibrated using an inverse fitting approach and is also shown in Fig. 13b. This calibration was performed by achieving the best fitting of the experimental and numerical force-

displacement curves for the SH, CH-R3, NDB-R8, and PS-R4 tensile tests. The calibrated damage- and fracture-related parameters are listed in Table 10.

To validate the DDI and DF loci as well as the calibrated damage evolution law, simulations were re-executed with the four different

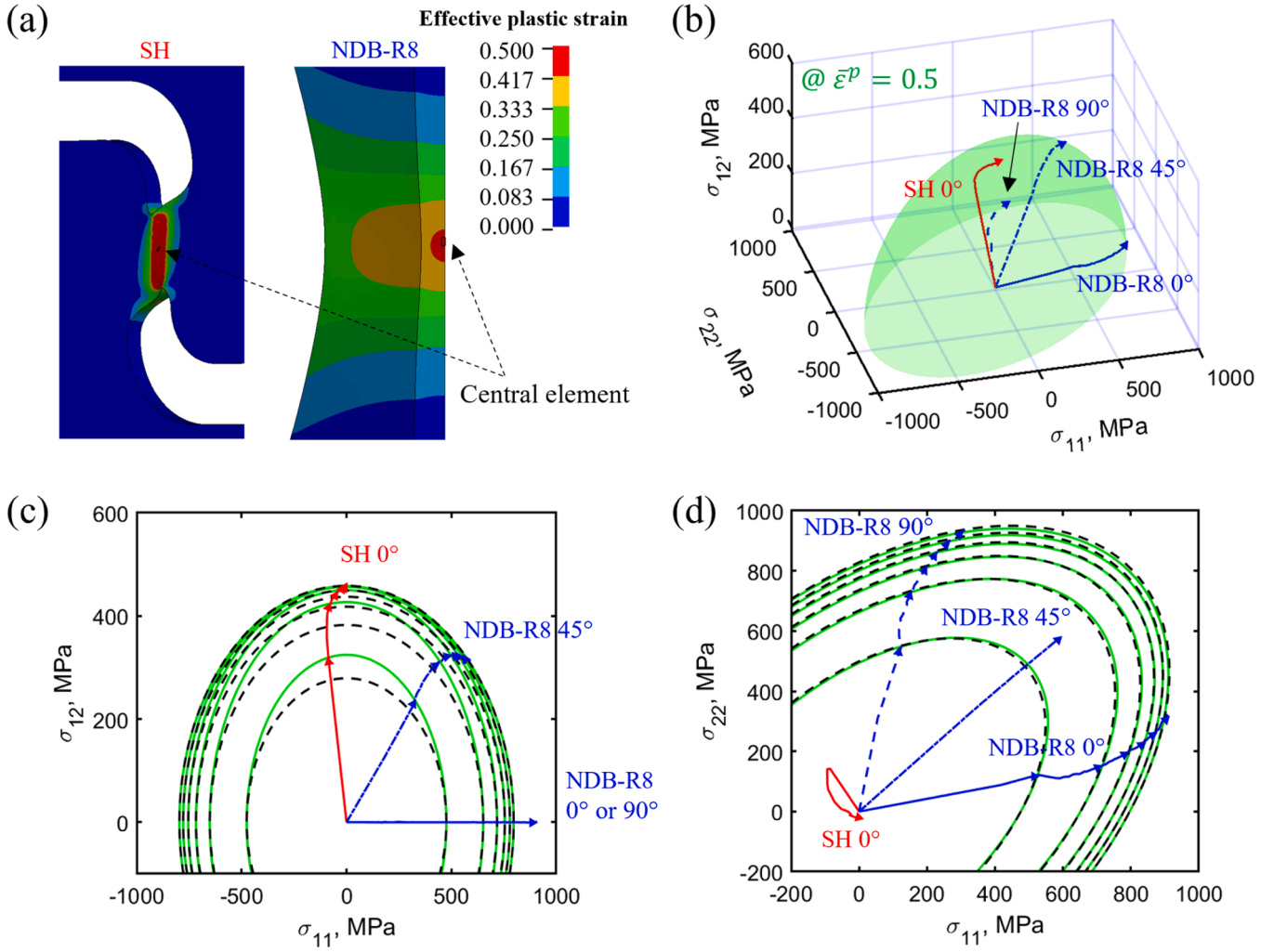


Fig. 18. YldSRH yield surface and the trajectory of the stress evolutions in simulations. (a) Strain contours of the SH and NDB-R8 specimens in the simulations. (b) YldSRH yield surface at a plastic strain of 0.5 and strain rate of 2000 s^{-1} (10000 mm/s) and the trajectory of the stress evolutions of the central elements of the SH and NDB-R8 specimens. (c-d) Stress evolution of the central elements in the $\sigma_{11} - \sigma_{22}$ and $\sigma_{11} - \sigma_{12}$ planes. The solid green and black dashed lines indicate the six yield loci of the YldSRH and Hill48 models at $\bar{\epsilon}^p = 0, 0.1, 0.2, 0.3, 0.4$, and 0.5 , respectively.

geometries under QS conditions. Fig. 13c-f shows the experimental and simulated force-displacement and surface local strain versus displacement curves at loading speed of 0.005 mm/s. The experimental surface local strain was evaluated from the center of necking zone of specimen using DIC. It can be recognized that good agreements in the overall experimental and simulated force-displacement response of four different specimens were achieved. Notably, the local surface strain evolution, which were not used in the calibration process, also showed strong agreement.

Adiabatic temperature influences not only plasticity, as discussed in Section 4.1.2, but also fracture behavior. To determine the effects of temperature on the fracture strain, uniaxial tensile tests were performed on NDB-R8 specimens at different temperatures (20°, 100°C, and 200°C) at a QS loading speed (0.005 mm/s). Fig. 14a shows the force-displacement curves and surface local strain-displacement curves of NDB-R8. A J-C temperature-dependent term $1 + F_5^{df} \cdot \frac{T - T_0}{T_m - T_0}$ (see also Eq. (24)) was used to fit the effects of temperature on fracture strain, as shown in Fig. 14b. It is found that the surface fracture strain measured using DIC decreased with increasing temperature. The value of F_5 is also listed in Table 10.

Ultimately, the parameter χ in Eq. (24), which controls the strain rate effect for different stress states, was calibrated by comparing the experimental and numerical force-displacement curves. Fig. 15 shows

the $\chi(\eta, \bar{\theta})$ value for the different specimen geometries with the corresponding stress triaxialities that can be fitted with a hyperbolic tangent function: $\chi(\eta) = \chi_1 \cdot \tanh(\chi_2 \cdot \eta + \chi_3) + \chi_4$. The calibrated parameters of this function are listed in Table 10. It can be seen in Fig. 15 that the strain rate sensitivity of SH to fracture strain, which is negative, is significantly higher than the slightly positive effects observed for CH-R3, NDB-R8 and PS-R4.

4.3. Validation of the YldSRH+e²MBW model at different loading speeds and directions

With the YldSRH+e²MBW model calibrated in Sections 4.1 and 4.2, FE simulations were performed for the SH, CH-R3, NDB-R8, and PS-R4 specimens at three orientations (0°, 45°, and 90°) at three loading speeds (0.005, 200, and 10000 mm/s) to evaluate the prediction precision of the new model.

Fig. 16 and Fig. 17 show the experimental and numerical force-displacement curves of the SH, NDB-R8, CH-R3, and PS-R4 specimens at the three loading speeds in both the YldSRH+e²MBW and Hill48+e²MBW models. The Hill48+e²MBW model only consider strain rate sensitivity at 0°, as shown in Fig. 12. It can be seen from Figs. 16 and 17 that at 0.005 mm/s, the simulations of both models and the experiments are identical, whereas slight differences appear at 200 mm/s. However,

at a high loading speed of 10000 mm/s, significant differences appeared between the two models, especially for the SH, 45° NDB-R8, CH-R3, and PS-R4 specimens. As shown in Fig. 16a-c, the YldSRH+e²MBW model demonstrates higher coincidence to the experimental results at a loading speed of 10000 mm/s for the SH specimen than does the Hill48+e²MBW model, which features isotropic strain rate hardening and excludes SH tests in its parameter determination (see Appendix C). For the NDB-R8, CH-R3 and PS-R4 specimens, both the Hill48+e²MBW and YldSRH+e²MBW models performed similarly and accurately captured the material dynamic plasticity under 0° and 90° tensile conditions (see Figs. 16d,f and 17a,c,d,f). However, as shown in Figs. 16e, 17b, and 17e, the Hill48+e²MBW model underestimated plasticity for the 45° NDB-R8, CH-R3, and PS-R4 specimens at a loading speed of 10000 mm/s. This drawback is addressed by the YldSRH+e²MBW model, which accounts for non-uniform strain rate hardening, resulting in improved predictions.

It is worth noting that in addition to plasticity, the YldSRH+e²MBW model also exhibits better predictions of the fracture initiation and subsequent fracture propagation compared to the Hill48+e²MBW model, especially for the SH and CH-R3 specimens under 200 and 10000 mm/s loading conditions, as shown in Figs. 16a-c and 17a-c. For the NDB-R8 and PS-R4 specimens, both models show similar predictions for material fracture points.

To explain the significant differences between the two models in predicting plasticity under high-speed loading, especially at 10000 mm/s, the stress evolution of the central elements of SH at 0° and NDB-R8 at 0°, 45°, and 90° (see Fig. 18a) during deformation until a plastic strain of 0.5 were extracted from the FE simulations, and the trajectories are plotted in the $\sigma_{11} - \sigma_{22} - \sigma_{12}$ stress space in Fig. 18b. The yield surface in Fig. 18b was obtained at a plastic strain of 0.5 under a strain rate of 2000 s⁻¹. This strain rate is close to the average strain rate of the local deformation zone of the SH and NDB-R8 specimens at a loading speed of 10000 mm/s.

For better visualization, the stress evolution trajectories are presented in the $\sigma_{11} - \sigma_{12}$ and $\sigma_{11} - \sigma_{22}$ planes (see Fig. 18c-d). The solid green lines depict the YldSRH yield locus at plastic strains ranging from 0 to 0.5, with a strain step of 0.1 between each locus. The black dashed line indicates the Hill48 yield loci with isotropic strain rate hardening. As can be seen in Fig. 18c, a high σ_{12} value of more than 400 MPa is observed for the 0° SH test, whereas in Fig. 18d, the corresponding σ_{11} and σ_{22} values are minimal. This confirms that the applied SH geometry (see Fig. 1b) achieved a relatively ideal simple shear stress state. Owing to the higher strain rate sensitivity for 0° SH in the YldSRH model than in the Hill48 model, the flow stress for the 45° NDB-R8 predicted by the YldSRH model was slightly higher than that predicted by the Hill48 model, as shown in Fig. 18c. This explains why the force on the 45° NDB-R8 specimen at a loading speed of 10000 mm/s in the YldSRH+e²MBW was higher than that in the Hill48+e²MBW model (see Fig. 16e). By contrast, for the 0° and 90° NDB-R8 tests, the shear stress σ_{12} was 0, and σ_{11} and σ_{22} approached 900 MPa at $\bar{\epsilon}^p = 0.5$ in the 0° or 90° direction (see Fig. 18d). As both the Hill48 and YldSRH models had almost identical yield loci at the $\sigma_{11} - \sigma_{22}$ plane, the simulation results using the Hill48+e²MBW and YldSRH+e²MBW were similar for 0° and 90° NDB-R8 tests, as shown in Fig. 16d and f. A similar comparison of results was also observed for the CH-R3 and PS-R4 specimens, as shown in Fig. 17, and an interpretation based on the NDB-R8 can also be applied.

5. Conclusion

In this work, tensile tests were conducted on H340 steel sheets with various specimen geometries, corresponding to different stress states (η , $\bar{\theta}$), at three material orientations (0°, 45°, and 90° to the RD) across strain rates ranging from 0.001 to 1000 s⁻¹. The strain rate sensitivities

of H340 were determined through uniaxial tensile and simple shear tests based on the oscillation-free forces measured at strain rates of up to 1000 s⁻¹. The results revealed that H340 exhibited significant strain rate hardening, with a pronounced loading direction-dependence in stress space. Specifically, the strain rate sensitivity was higher in shear at 0° compared to uniaxial tension in all directions. Additionally, tension at 45° showed greater sensitivity than both 0° and 90° uniaxial tensions.

The loading direction- and stress state-dependent strain rate hardening resulted in a non-uniform expansion of the yield surface with increasing plastic strain, where the yield surface shapes were also strain rate-dependent. To accurately describe this non-uniform strain rate hardening behavior, a new non-uniform strain rate dependent yield surface model (YldSRH) based on the Tsai-Wu model was developed. The YldSRH model features independent parameters that can be directly determined from experimental data, enhancing both its practical applicability and predictive accuracy.

The proposed YldSRH model, coupled with a rate- and temperature-dependent damage mechanics model (e²MBW), achieved good agreement between the experimental and numerical force-displacement responses for the SH, CH-R3, NDB-R8, and PS-R4 specimens across loading speeds and material orientations. The YldSRH+e²MBW model exhibited significantly improved predictability of the anisotropic plasticity and fracture behavior of H340 compared with the Hill48+e²MBW model, particularly under SH and CH-R3 loading cases, and high-speed loading conditions (10000 mm/s).

Although some simplifications were made, such as averaging to calculate the equi-biaxial tensile strength using $B_{t,xy} = (U_{t,0^\circ} + 2U_{t,45^\circ} + U_{t,90^\circ})/4$, and the assumption that the uniaxial tensile and compressive yield stress are identical, the YldSRH model successfully captured the key experimental trends. This was particularly evident in the σ_{12} direction, where the yield surface expands non-uniformly with increasing strain rate, distinguishing it from the Hill48 model.

In the future, high-speed uniaxial compression and biaxial tension and compression tests on bar materials, or simulations of out-of-plane tests using a crystal plasticity model, are planned for further validation and enhancement of the YldSRH yield surface. The proposed model can also be extended to a broader range of anisotropic materials under dynamic or impact loading conditions.

CRedit authorship contribution statement

Chongyang Zeng: Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Xiangfan Fang:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Xiangfan Fang reports financial support was provided by University of Siegen. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Shear stress and equivalent strain determination for simple shear tests

As proposed in previous studies [71,106], the gauge area of the simple shear test remained almost constant during the shear test using SH geometry (see Fig. 1). Thus, the average shear stress, τ_{avg} , could be calculated as follows:

$$\tau_{avg} = \frac{F}{A} = \frac{F}{Lt}, \quad (A1)$$

where F is the measured force, L is the length of the shear region, and t is the sheet thickness.

By assuming that normal stresses are negligible in simple shear, the work-conjugate equivalent strain [70,72] was derived by

$$\varepsilon_{eq}^p = \left(\frac{\sigma_{12}}{\sigma_{eq}} \right) \gamma = 2 \left(\frac{\sigma_{12}}{\sigma_{eq}} \right) \sinh(\varepsilon_1), \quad (A2)$$

where ε_1 is the major strain that can be determined using the DIC method. γ is used to define the engineering shear strain. σ_{12}/σ_{eq} is the shear stress ratio, which was assumed to be constant in this study. It is worth to note that the shear stress ratio may be valid only for a specific range of strain for materials that have significant evolving anisotropy. Rahmaan et al. [71] proposed a criterion to estimate the shear stress ratio using a non-quadratic Hosford yield criterion

$$\frac{\sigma_{12}}{\sigma_{eq}} = \left(\frac{1}{2^{a-1} + 1} \right)^{1/a}, \quad (A3)$$

where $a = 2$ corresponds to the von Mises yield criterion, $a = 6$ is recommended for BCC materials, and $a = 8$ is recommended for FCC materials. The resulting shear stress ratios for $a = 2$, $a = 6$, and $a = 8$ are approximately 0.577, 0.558, and 0.545, respectively. As the investigated material, H340, is ferritic (BCC) steel, 0.558 was used to calculate the shear stress ratio in this study.

Appendix B. Parameter calibration for the Tsai-Wu failure criterion

By applying a uniaxial tensile stress to a specimen in the direction of the x-axis direction, tensile damage can be measured, and the corresponding failure stress is designated as X . Similarly, the uniaxial compressive strength along the same axis can be experimentally determined and is denoted as X' .

Let $i = 1$ such that Eq. (6) takes the form:

$$F_1 \sigma_{11} + F_{11} \sigma_{11}^2 = 1 \quad (B1)$$

When $\sigma_{11} = X$, or $\sigma_{11} = -X$, Eq. (B1) becomes:

$$\begin{cases} F_1 X + F_{11} X^2 = 1 \\ -F_1 X' + F_{11} X'^2 = 1 \end{cases} \quad (B2)$$

From Eq. (B2), the following F_1 and F_{11} expressions are derived:

$$F_1 = \frac{1}{X} - \frac{1}{X'}, \quad F_{11} = \frac{1}{XX'} \quad (B3)$$

Similarly, through uniaxial tensile and compressive tests conducted along the y- and z-axis, the following relations are obtained:

$$F_2 = \frac{1}{Y} - \frac{1}{Y'}, \quad F_{22} = \frac{1}{YY'}, \quad F_3 = \frac{1}{Z} - \frac{1}{Z'}, \quad F_{33} = \frac{1}{ZZ'}, \quad (B4)$$

where Y and Y' represent the uniaxial tensile and compressive strengths along the y-axis, and Z and Z' represent those along the z-axis.

By conducting a pure shear test in the three orthogonal planes, the following shear strength relations are derived:

$$F_{44} = \frac{1}{QQ'}, \quad F_{55} = \frac{1}{RR'}, \quad F_{66} = \frac{1}{SS'}, \quad (B5)$$

where Q and Q' are the positive and negative pure shear strengths along the x-y plane; R and R' along the y-z plane; and S and S' along the x-z plane.

If a biaxial tension is performed such that:

$$\sigma_{11} = \sigma_{22} = B_{t,xy}, \quad \sigma_{33} = \sigma_{12} = \sigma_{23} = \sigma_{13} = 0 \quad (B6)$$

Substituting these stresses in Eq. (6) will result in the following expression:

$$B_{t,xy}^2 (F_{11} + F_{22} + 2F_{12}) + B_{t,xy} (F_1 + F_2) = 1 \quad (B7)$$

Solving for F_{12} results in:

$$F_{12} = \frac{1}{2B_{t,xy}^2} \left[1 - B_{t,xy} (F_1 + F_2) - B_{t,xy}^2 (F_{11} + F_{22}) \right] \quad (B8)$$

In addition, F_{12} can also be calculated based on the 45° tensile strength $U_{t,45^\circ}$ or the 45° off-axis pure shear strength S_{45° , as shown in Eq. (B9). A more detailed parameter definition and calibration process can be found in a previous study [75].

$$F_{12} = \frac{2}{U_{t,45}^2} \left[1 - \frac{U_{t,45}^2}{4} (F_{11} + F_{22} + F_{44}) \right]$$

$$F_{12} = -\frac{1}{2S_{45}^2} [1 - S_{45}^2 (F_{11} + F_{22})] \quad (B9)$$

Appendix C. Hill48 yield criterion and calibration of the corresponding parameters

Hill [1] proposed one of the most widely used anisotropic yield criteria, known as Hill48, in 1948. The Hill48 yield function is defined as follows:

$$\bar{\sigma}(\sigma)_{Hill48} = \sqrt{\frac{1}{2} [F_{\sigma}(\sigma_{22} - \sigma_{33})^2 + G_{\sigma}(\sigma_{33} - \sigma_{11})^2 + H_{\sigma}(\sigma_{11} - \sigma_{22})^2] + L_{\sigma}\sigma_{23}^2 + M_{\sigma}\sigma_{13}^2 + N_{\sigma}\sigma_{12}^2} \quad (C1)$$

where F , G , H , L , M , and N are six anisotropic parameters, usually calibrated using data from uniaxial or biaxial tensile tests. The calibration of these parameters for the Hill48 yield function ($L_{\sigma} = M_{\sigma} = 3$ for the case where out-of-plane anisotropy is not considered) is given in the following equations [19,74,80,107].

$$F_{\sigma} = \frac{\sigma_0^2}{\sigma_{90}^2} - 1 + \frac{\sigma_0^2}{\sigma_b^2}, \quad G_{\sigma} = 1 - \frac{\sigma_0^2}{\sigma_{90}^2} + \frac{\sigma_0^2}{\sigma_b^2}, \quad H_{\sigma} = 1 + \frac{\sigma_0^2}{\sigma_{90}^2} - \frac{\sigma_0^2}{\sigma_b^2}, \quad N_{\sigma} = \frac{4\sigma_0^2}{\sigma_{45}^2} - \frac{\sigma_0^2}{\sigma_b^2} \quad (C2)$$

Data availability

Data will be made available on request.

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